DATA SYSTEMS

PosPro

GNSS/INS trajectory post-processing

STAGE 01 OF THE DCS CHAIN · POSPRO → ALIGNPRO → CLOUDPRO

User Guide

For PosPro 2.4.0 · November 2026

Canadian Lidar · QGIS plugin, part of the Data Collection Systems chain. Needs the DCS Core plugin.

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1 Introduction

PosPro computes the trajectory of a Canadian Lidar mobile mapper: where the Certus GNSS/INS was, and how it was oriented, at every moment of a drive. It reads the mapper's captures (`.clraw`) or the Certus logs (`.anpp`) directly, combines the raw IMU and the raw satellite observations of both Certus antennas with base station data, and runs the inertial filter forward and backward over the drive. The two passes are combined into one smoothed trajectory,

which you check on the map and in plots, clean up in Trajectory Review and export as SBET/SMRMSG, text, GeoPackage or GPX.

PosPro is **stage 01** of the DCS chain. Every time you process, PosPro also finalizes the trajectory for the next stages: **AlignPro** uses it to solve the Lidar mounting (boresight), and **CloudPro** uses it to georeference the point cloud. You do not export anything by hand for them.

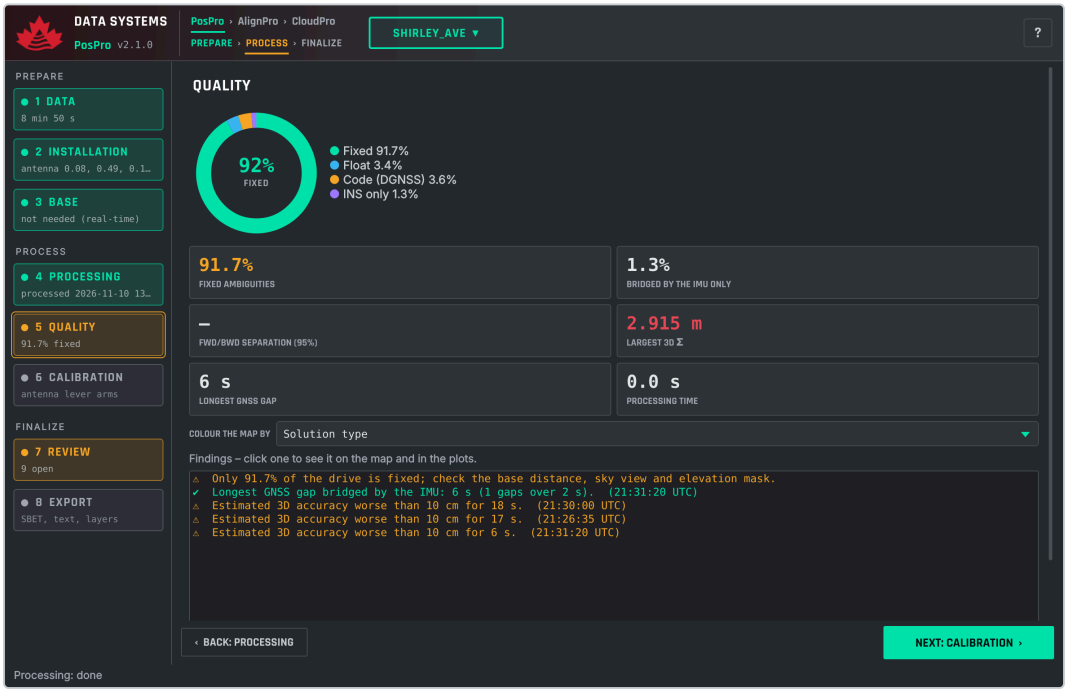


Figure 1.1 The PosPro window: the header with the chain, the phases and the project box, the eight steps in three phases (Prepare, Process, Finalize) on the left, the open step on the right, Back and Next under it and the status line at the bottom.

What PosPro reads and makes

	DETAILS
Rover data	Mapper captures (`.clraw`, also unfinished `.clraw.partial` files) and Certus logs (`.anpp`). PosPro uses the Certus packets 20 (real-time solution and attitude), 28 (raw IMU), 60 (raw satellite observations), 61 (GPS ephemeris), 185 (installation) and 196 (dual-antenna set-up), and the mapper's `hardware.json` inside the capture.
Base stations	RINEX observation files (also Hatanaka-compressed and gzipped), or downloads from the base-station finder. Base positions from the RINEX header, typed in, built-in PPP or NRCan CSRS-PPP.
Processing	Tightly coupled GNSS/INS (both antennas), loosely coupled, GNSS only, or the Certus real-time solution. Differential (PPK) or precise point positioning (PPP). Forward and backward passes, combined.
Checks	Quality summary and findings, map layers coloured by solution type, accuracy or separation, quality plots, Trajectory Review, antenna lever-arm solver.
Output	SBET + SMRMSG, text files with the columns you choose, GeoPackage layer, GPX track, export note; the finalized trajectory for AlignPro and CloudPro.

Processing runs in a separate engine program (with RTKLIB inside), so a fault in processing never closes QGIS. The forward and backward passes run at the same time.

How to use this guide

Read *How a trajectory is made* and *The workflow and why it is in this order* once: they explain what every setting does to the accuracy. *Best practices* lists what to do in the field and in the office. The step chapters then follow the window,

step by step; each starts with why the step comes where it does. The Index at the end lists every button and setting with the section that explains it.

> NEW IN 2.4.0 The steps are in a new order, grouped in three phases: Installation now comes before Base, and Calibration before Review (see *The workflow*). Full list: Version history.

2 How a trajectory is made

A trajectory is the position and orientation of the Certus at every moment of the drive. PosPro builds it from two kinds of measurement that fail in opposite ways, and from what you tell it about the vehicle. Knowing how they fit together explains every setting in this guide and why the steps come in the order they do.

Two sensors that cover for each other

	GNSS (THE TWO ANTENNAS)	IMU (INSIDE THE CERTUS)
Measures	Distances to satellites: code (metres) and carrier phase (millimetres, once its whole number of cycles, the *ambiguity*, is known)	Acceleration and turn rate, 200 times a second
Strength	Errors do not grow with time: centimetres anywhere the sky is open	Smooth, never drops out, gives roll, pitch and heading
Weakness	Blocked by bridges, trees, buildings; carrier phase must be fixed again after every interruption	Errors grow with time when nothing corrects them (metres after a minute)

The **Kalman filter** combines them: the IMU carries the trajectory from one moment to the next, and every GNSS measurement pulls it back to where the satellites say it is. In **tight coupling** (PosPro's default) the filter uses each satellite's raw observations from both antennas directly, so even two or three satellites under trees still help; loose coupling only uses complete GNSS positions.

Differential: the base station

Most GNSS errors (satellite orbits and clocks, the ionosphere and troposphere) are almost the same for two receivers a few kilometres apart. Processing the rover's observations *against* a base station's (double differences, PPK) removes them and makes the carrier-phase ambiguities fixable: that is what turns metres into centimetres. Two consequences:

- The whole trajectory sits on the **base station's position**: an error there moves every point by the same amount. That is why the base position is computed carefully (PPP or CSRS-PPP), not taken from the RINEX header.
- Accuracy falls with **distance to the base**, about 1 cm + 1 mm per km (1 ppm), and fixing gets harder beyond 20–30 km.

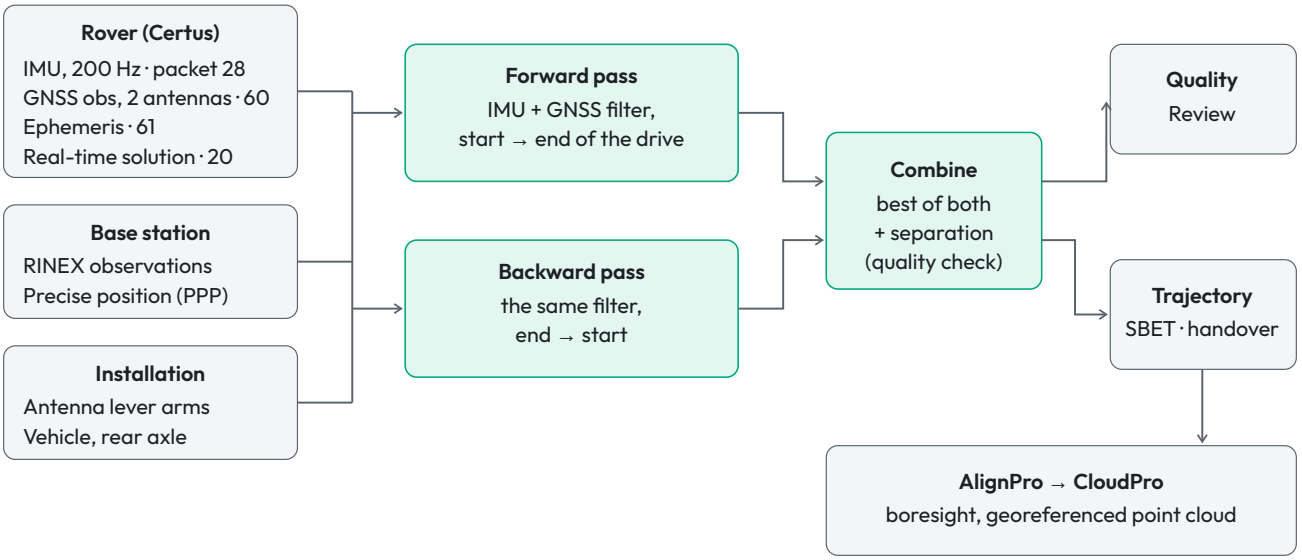


Figure 2.1 What goes into a trajectory and where it goes. The three inputs on the left are the three *Prepare* steps; the filter, the combination and the checks are *Process*; the trajectory and what uses it are *Finalize*.

Forward, backward and combined

A filter only knows the past. Driving forward, it enters a GNSS gap well known and leaves it with its largest error; running the same data backward, the opposite. Combining the two passes (weighting each by its own accuracy) gives the smallest error everywhere, and halfway through a gap it is much better than either. The **separation** between the passes is also the most honest quality check there is: two independent solutions that agree to 2 cm are right to about 2 cm; where they disagree, one of them has a wrong fix.

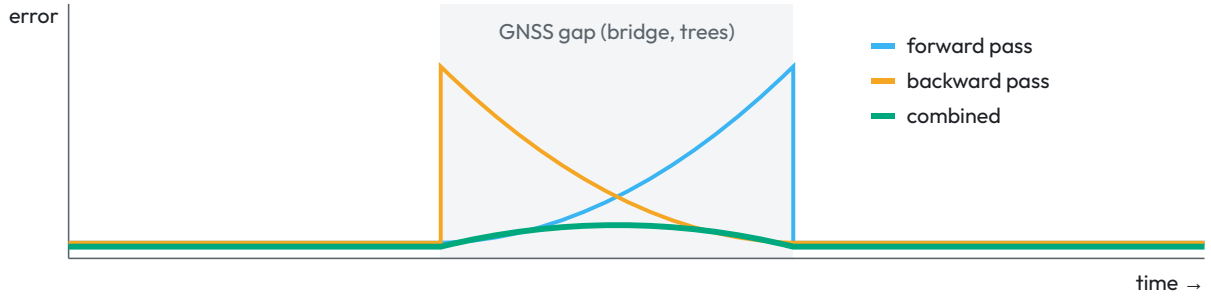


Figure 2.2 Through a GNSS gap each pass is best where it enters and worst where it leaves; the combined trajectory is the best of both.

What limits the accuracy

Each item below adds to the final error. The order of the steps is the order in which they are removed:

SOURCE	TYPICAL SIZE IF IGNORED	REMOVED IN
Antenna lever arms (where the antennas are, from the Certus)	The full error of the offset, in every direction the vehicle turns; swapped antennas: about 1 m	Installation (measured, checked), Calibration (solved)
Base station position	Up to a few metres, the same everywhere	Base (PPP or CSRS-PPP)
Distance to the base, atmosphere	1 mm per km, more in disturbed ionosphere	Base (choose the closest), Processing (atmosphere states)

Wrong or unfixed ambiguities	Decimetres to metres	Processing (tight coupling, both antennas, both passes), Quality, Review
GNSS gaps (IMU only)	Grows with the gap: centimetres for seconds, decimetres after a minute	Processing (combination), Review (bridge with the IMU)
Vehicle frame (IMU mounting, wheel constraints)	Small heading and pitch biases, worse in gaps	Installation (rear axle, IMU mounting correction)

3 The workflow and why it is in this order

PosPro's eight steps are grouped in three phases. Each step depends on the ones before it, and the step bar on the left, the `NEXT` button and the step notes guide you through them in order.

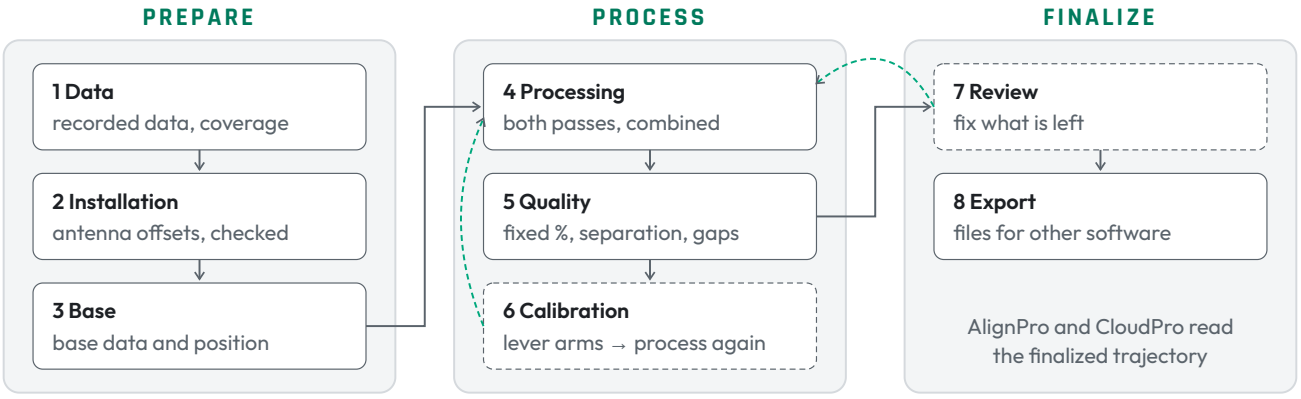


Figure 3.1 The three phases. Solving the lever arms (Calibration) and bridging with the IMU (Review) both send you back to Processing, which is why they come after it, and why Calibration comes before Review.

In the window, the steps sit on the left under their phase. Each shows its state in colour with a short note under its name:

- **Green**: done (for example *99.8% fixed*, *3 reviewed*, *not needed (real-time)*).
- **Amber**: needs a look (a base position from the RINEX header, an antenna offset of zero, review windows still open, fewer than 95 % fixed).
- **Red**: a problem to fix before going on.
- **Grey**: not done yet; the note says what is missing.

Under the open step, the `‹ BACK` / `NEXT ›` bar says in one sentence whether the step is done and what to press; optional steps can be skipped.

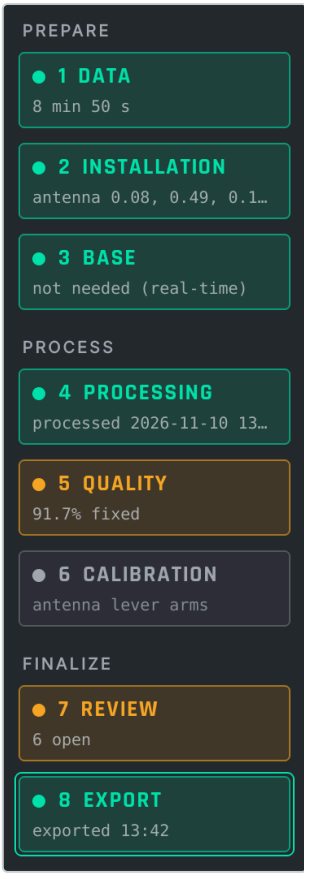


Figure 3.2 The steps, in their phases, with their states.

STEP	WHY IT COMES HERE

1 Data	Everything else is computed from the rover files. Reading them first shows how complete they are (IMU and satellite coverage, gaps in the recording) before any time is spent on them.
2 Installation	The antenna offsets are used in every epoch of processing, and a wrong offset (or swapped antennas) cannot be fixed afterwards without processing again. They come straight from the capture, so this is mostly a check, and `CHECK ANTENNAS` confirms them from the data itself.
3 Base	The base position decides where the whole trajectory sits. It is computed (PPP) before processing, because every PPK solution is relative to it.
4 Processing	Runs once everything it needs is right.
5 Quality	Judges the result before anything is built on it: the share of fixed ambiguities, the forward/backward separation and the gaps say whether to go on, change a setting, or check the inputs.
6 Calibration	Solving the lever arms needs a processed drive and means processing again. Done after a first good processing and before any review, it leaves no hand edits to lose.
7 Review	Hand edits belong to one processed trajectory, so they come last; processing again takes hand selections off the list.
8 Export	Only for other software: AlignPro and CloudPro already read the trajectory PosPro finalizes every time it processes or applies a review edit.

> `MOST JOBS` With good data the path is short: Data (read), Installation (check), Base (add, PPP), Processing (process), Quality (look), and on to AlignPro. Calibration is for a new installation; Review only where Quality points to a problem.

4 Best practices

Most of a trajectory's accuracy is decided before PosPro runs: by how the data were recorded. These are the practices used across the industry for GNSS/INS mobile mapping, and how PosPro supports each.

In the field

PRACTICE	WHY
Measure the antenna and Lidar offsets to the Certus reference point to 1 cm, and enter them in the mapper (*Settings › Hardware*)	Offsets go straight into every position. PosPro reads them from the capture; `CHECK ANTENNAS` confirms GPS1 and GPS2 from the data.
Check that PRIM and SEC are plugged into the right antennas	Swapped antennas move the trajectory by the antenna separation and turn the Certus's heading 180°.
Start and end with 3–5 minutes static in the open, engine running	Lets the filter settle its biases and fix the ambiguities before the drive, and gives the backward pass the same at the end.
Drive a few turns and figure-eights early on	Makes heading and the IMU biases observable; the lever-arm solver needs turns.
Log the base for the whole drive plus 10 minutes before and after, at 1 s; within 10–20 km where possible	The base must cover every rover epoch; accuracy falls about 1 mm per km; 1 s matches the rover's whole-second epochs.
Record the mapper's navigation recording (mapper 2.10), not only the survey lines	One continuous recording from start to end processes as one trajectory that covers every line, static periods included.
Use the mapper's recommended Certus outputs	Raw IMU at 200 Hz and raw observations of both antennas at 8 Hz on whole seconds are what tight coupling needs (see Recording settings).
Avoid long GNSS outages where you can; pass through them at a steady speed	IMU-only error grows quickly with time; the wheel constraints help most at constant speed.

In the office

PRACTICE	WHY
Compute the base position with CSRS-PPP (best, after about 24 h when final orbits exist) or built-in PPP; never leave the RINEX header position	The header position is often metres off; every point of the trajectory inherits it.
Use tight coupling with both antennas and both passes	The most observations, the best bridging of gaps, and the separation check.
Accept a trajectory when: fixed ≥ 95 %, forward/backward separation 95 % below 5 cm, no IMU-only gap longer than about 30 s	The common acceptance criteria for survey-grade mobile mapping; Quality shows each.
Fix inputs, not results: a bad base file, wrong offsets or a wrong setting are fixed and processed again before any review	Review edits are for what the data cannot give (a wrong fix under trees), not a substitute for correct inputs.
Solve the lever arms once per installation, from a drive with turns, and keep the measured values when the solver disagrees by more than a few centimetres without a reason	A solved value is only as good as the drive it came from.
Prefer bridging with the IMU over interpolation for anything longer than a few seconds	The IMU knows the turns and bumps; a curve guesses them.

Keep the export note with every exported trajectory	It records the frame, epoch, reference point and time base, which the next user needs.

5 Installing and opening PosPro

PosPro is one of three QGIS plugins that process a Canadian Lidar mobile mapper job one after the other. All three need the **DCS Core** plugin, which holds what they share: the job's processing project, the files each stage hands to the next, the look, the header and the window.

STAGE	PLUGIN	READS	FINALIZES
01	**PosPro**	The mapper captures (`.clraw` or `.anpp`), base station data	The trajectory: `trajectory.json` with SBET and SMRMSG
02	**AlignPro**	The captures and PosPro's trajectory	The Lidar mounting: `alignment.json`
03	**CloudPro**	The captures, the trajectory and the mounting	The point cloud: `cloud.json`, LAS 1.4, ground grid, report

Requirements

- QGIS 3.34 or later, including QGIS 4.
- DCS Core 1.14.0 or later. PosPro 2.4.0 will not start with an older DCS Core: a message bar asks you to update it.

Install from the Canadian Lidar repository

1. In QGIS open `PLUGINS › MANAGE AND INSTALL PLUGINS › SETTINGS`, press **Add...** and enter the repository URL `https://canadian-lidar.github.io/qgis-plugins/plugins.xml`.
2. Press **Reload all Repositories**, then install **DCS Core** first and **PosPro** after it.
3. Restart QGIS. Later versions appear under `UPGRADEABLE` in the same window; update DCS Core together with the plugins.

Without internet access, use `INSTALL FROM ZIP` in the same window with `dcs_core-1.14.0.zip` first, then the PosPro zip.

Opening PosPro

- `PLUGINS › CANADIAN LIDAR › DATA SYSTEMS › POSPRO`, or the PosPro button on the Plugins toolbar.
- From another DCS plugin: click the **PosPro** box in its header to open PosPro on the same project.
- This guide: the `?` button in the header, `USER GUIDE (THIS VERSION)` in the project box, or `HELP › PLUGINS › POSPRO USER GUIDE`. The guide that opens always matches the installed version.

6 The PosPro window

PosPro, AlignPro, CloudPro and RemixPro share the same header and the same kind of window, so what you learn here applies to all of them.



Figure 6.1 The header. Left: the logo, the plugin name and version, then two rows of breadcrumbs: the chain of stages and this plugin's own steps. Then the project box, and the user guide button at the right.

The header

PART	WHAT IT DOES
Plugin name and version `PosPro v2.1.0`	The plugin and the version you are running. Quote it when you report a problem; the user guide (`?`) is the one for this version.
The chain `PosPro › AlignPro › CloudPro`	The state of each stage on the open project, by colour: **green** finalized and current, **amber** out of date (an earlier stage was run again, the raw data or a shared setting changed), **blue** running, **red** failed, **grey** not run yet. The plugin you are in is underlined. Hover a name for the state and the reasons; click another stage to open that plugin on the same project. The chain refreshes every few seconds.
PosPro's steps `PREPARE › PROCESS › FINALIZE`	The three phases of the step list on the left: Prepare (Data, Installation, Base), Process (Processing, Quality, Calibration) and Finalize (Review, Export). Each step is coloured the same way (green done, amber needs a check, red needs attention, blue running, grey not done yet); the current one is underlined. Hover a step for what is done and what to do next; click it to go there.
Project box	The open project's name; click it for the project menu (Projects). It turns red while PosPro is working.
`?`	Opens this guide.

Moving through the steps

Under the step pages, `‹ BACK` goes to the previous step and `NEXT: <STEP> ›` to the next one; Next is greyed out until a required step is done and reads `SKIP TO <STEP> ›` on optional steps. Hover either button for what is done and what to do (the sentence that used to sit between them). On the last step Next becomes `CONTINUE IN ALIGNPRO ›`, which opens AlignPro on the same project once the trajectory is finalized. The step list on the left and the steps in the header still open any step at any time.

The window

PosPro opens as a **window of its own**, centred on the QGIS screen the first time. Use the system's title bar to move it, maximise, minimise or close it; it does not dock into the QGIS window. PosPro remembers its place and size, or that it was maximised, and opens the same way next time, also after QGIS restarts. `RESET WINDOW` in the project menu brings it back to its default size, centred.

Below the steps, the bottom of the window holds the **status line** (the last message, or the stage of the job that is running), a progress bar and a red `STOP` button while a job runs, and short **messages** (green, amber or red edge) that disappear after a few seconds. Warnings and errors also go to the QGIS message bar and to `SHOW THE LOG`. Only one job runs at a time; starting another while one runs gives *Wait for the current job to finish (or cancel it).*

The default window is 1400 × 950 pixels.

7 Projects and the job folder

PosPro, AlignPro and CloudPro work on one shared **processing project** per job. The mapper's capture folders are never changed: everything the plugins make goes in a `Processing` folder.

PATH	HOLDS
`Job_Oct20/`	The job folder: the capture folders recorded by the mapper (read only)
`Job_Oct20/Processing/<Name>/<Name>.dcsproj`	The shared project: captures, shared settings (coordinate system, base station, antennas, Lidar mounting, vehicle) and the state of each stage. A `.bak` copy is kept beside it.
`01_PosPro/`, `02_AlignPro/`, `03_CloudPro/`	Each stage's own settings file, `work/`, `log/`, and `out/` with what it finalizes
`<Name>.qgz`	The QGIS project, when you save it from the project menu

Mapper 2.10 projects

From mapper 2.10 the mapper keeps its own projects. Use the mapper project's folder as the job folder:

PATH	HOLDS
`Deck_Test/project.json`	The mapper project: name, id, next line, operator, vehicle
`Deck_Test/lines/Deck_Test_L001_20261104_124512/`	One survey line (one Start/Stop), its files named `Deck_Test_L001_20261104_124512_0001.clraw`, `_0002` … Captures made before 2.10 that the mapper grouped into the project keep their old names here.
`Deck_Test/nav/Deck_Test_NAV_20261104_120000_0001.clraw`	A navigation recording: everything the Certus sent while the project was open (no Lidar). A new one starts at each power-on, project opening, UTC date change and every 2 GB.

If you pick the project's `lines` or `nav` folder, or a line folder, the New project dialog takes the mapper project's folder as the job folder. The survey lines are recognised from what the files record (their line number and capture), not from their names, so a line split over several files is one line and old and new captures mix freely, in recording order. Navigation recordings are never taken as survey lines.

The Processing folder can also live on another drive (choose **Elsewhere…** when you create the project). Paths are stored relative to the job folder where possible, so a job can be copied or moved as a whole.

The project menu

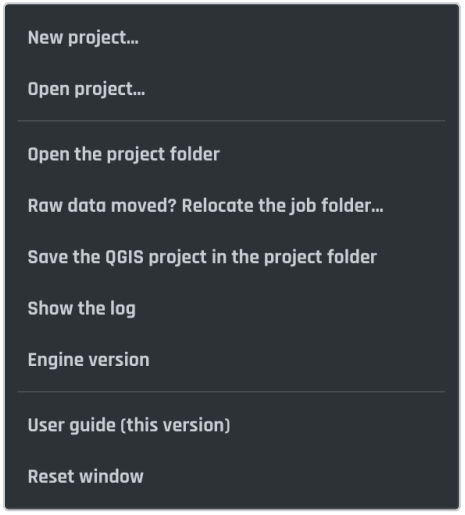


Figure 7.1 The project box menu in PosPro.

ITEM	WHAT IT DOES
NEW PROJECT…	Opens the New DCS project dialog (below). PosPro then offers the captures it finds in the job folder as rover files.
OPEN PROJECT…	Opens a processing project (`.dcsproj`), or an older stand-alone PosPro project (`.pospro.json`). The last project you had open reopens by itself when PosPro starts.
OPEN THE PROJECT FOLDER	Shows the project folder (`Processing/<Name>`) in the file manager.
RAW DATA MOVED? RELOCATE THE JOB FOLDER…	Asks for the job folder's new location and finds the captures there again; PosPro's rover file list follows. Not available for older stand-alone projects.
SAVE THE QGIS PROJECT IN THE PROJECT FOLDER	Saves the open QGIS project as `<Name>.qgz` in the project folder (asks first when the QGIS project already has another file name).
SHOW THE LOG	The PosPro log of this session: every message, warning and error with its time.
ENGINE VERSION	The version of the processing engine and of RTKLIB inside it, and where the engine is installed.
USER GUIDE (THIS VERSION)	Opens this guide.
RESET WINDOW	The window back to its default size, centred.

Creating a project

1. Choose NEW PROJECT… in the project box.
2. **Job folder**: the folder that holds the mapper's capture folders. The project name defaults to the date of the first capture with a number, for example *2026-11-03_1*; when that project exists already the next number is offered (*2026-11-03_2*), so a second processing of the same job (another base station, other settings) gets its own folder. Type any other name if you prefer.
3. **Processing folder**: inside the job folder by default; **Elsewhere…** to put it on another drive.
4. Press OK. The project is created and opened. PosPro then looks through the job folder (five folder levels deep, skipping the Processing folder and the mapper's processed folders) for mapper captures: the survey lines'

`.clraw` files and the mapper 2.10 navigation recordings, or the Certus logs (`.anpp`) when there is no `.clraw`, and asks whether to use them as the rover files. When a mapper 2.10 project holds both navigation recordings and survey lines, it asks which (Figure 7.3).

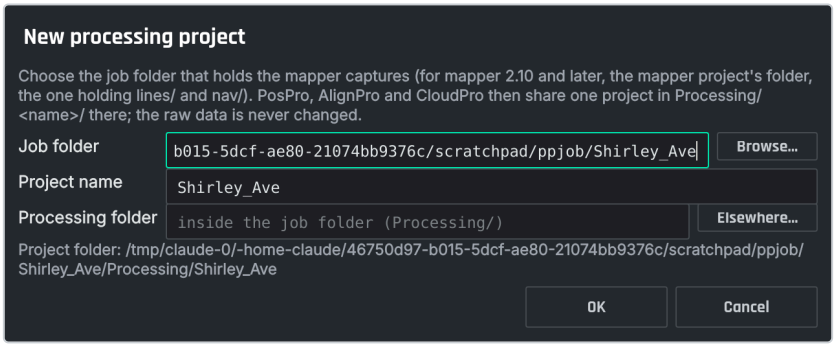


Figure 7.2 The New DCS project dialog (the same in every plugin).

A project made in any of the three plugins is the same project: open the `.dcsproj` in another plugin, or click its box in the header.

When the raw data has moved

If the captures are not where they were (another drive letter, a copied job), PosPro says so when the project opens and asks for the job folder's new location; **RAW DATA MOVED? RELOCATE THE JOB FOLDER…** in the project menu does the same at any time. The captures are found again by their place in the job folder; nothing goes out of date because of the move.

Out-of-date results

Each stage records what it was built from. When an earlier stage finalizes again, the raw data changes, or a shared setting it depends on is edited, the stage turns **amber (out of date)** in the chain in the header, with the reason on hover. Run that stage again to bring it up to date. A run that fails or is cancelled keeps the earlier results.

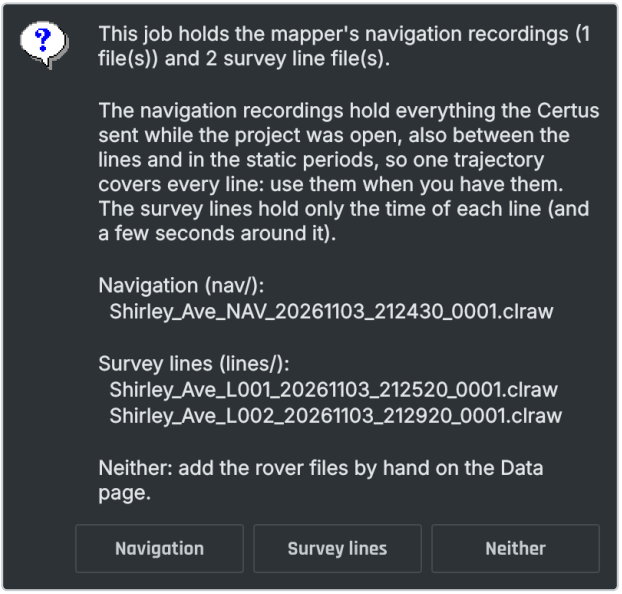


Figure 7.3 A mapper 2.10 project with a navigation recording and survey lines: choose **NAVIGATION** (recommended), **SURVEY LINES** or **NEITHER**.

Navigation or survey lines? A mapper 2.10 navigation recording holds everything the Certus sent while the mapper project was open: the static periods before and after the drive and the time between the lines as well as the lines themselves. Processed once, it gives one trajectory that covers every line, which suits PPK and tight coupling best. The survey lines hold only each line and a few seconds around it. Files are read in recording order, whatever their names. The navigation recording's hardware record has no vehicle, rear axle or coordinate system; PosPro takes those from a survey line of the same job.

PosPro keeps its own settings in `01_PosPro/pospro.json` and its work in `01_PosPro/work/` (extracted data, intermediate files, results). Shared facts that the other stages need (coordinate frame, base station, antenna offsets, vehicle, the capture's hardware record, the list of captures) are written into the shared project whenever you change them in PosPro.

8 Step 1 - Data

> **WHY HERE** Everything is computed from the rover files. Reading them first shows how complete the recording is (IMU and satellite coverage, gaps) before any time is spent on it; a recording with holes is better found now than after processing.

The Data page holds the rover files, what they contain, the time range to process and the satellite ephemeris.

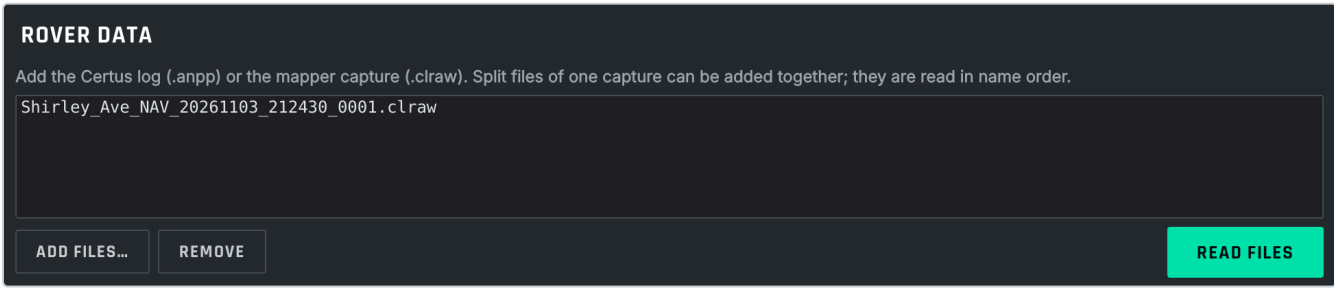


Figure 8.1 The rover files of the project: the two captures found in the job folder.

Rover data

- **ADD FILES…** adds Certus logs (`.anpp`) or mapper captures (`.clraw`, `.clraw.partial`). Split files of one capture can be added together; they are read in name order. Adding files clears the time range and reads the files at once.
- **REMOVE** takes the selected files off the list (the files themselves are not touched).
- **READ FILES** reads all files again. Reading extracts the raw IMU, the raw observations of both antennas, the real-time solution, the ephemeris and the capture's hardware record into the project's work folder. PosPro reads again by itself before processing whenever a file has changed.

A file that is no longer where it was shows in red (*File not found* on hover). After reading, PosPro fills the Installation offsets from the capture if they are still empty, with the message *Installation offsets filled from the capture (…)*.

What the files contain

After reading, this box lists what PosPro found. Lines in amber point at something to check.

LINE	SHOWS
Recording	Start and end (UTC) and duration of the IMU data; for captures without raw IMU, of the Certus real-time solution (the line says so).
IMU (packet 28)	Number of samples, rate and gaps (amber when there are gaps). Notes when samples before GNSS time was valid were skipped, or when IMU time stamps were shifted to match packet 20. When 2 % or more of the IMU data is missing, a warning gives the share and the longest hole: the recording lost Certus data, and inertial results will be poorer than the data allow until the mapper's link to the Certus is fixed.
GNSS (packet 60)	Epochs, interval and the mean number of satellites, then the signals recorded. Amber notes: epochs not on whole seconds (base data are then interpolated; set the Certus to 125 ms for whole-second epochs), or GLONASS off on the Certus. *none – raw observations were not recorded* when the captures hold no raw observations; the Data step then turns red (*no raw GNSS*) and only the real-time mode can run.
Ephemeris (packet 61)	The number of GPS ephemerides the Certus recorded.

Checksum errors	Shown only when packets failed their checksum.
Mapper hardware	*hardware.json found in the capture*: the mapper's offsets can be used on the Installation page.

Time range to process (UTC)

`START` and `END` limit the processing to part of the capture, for example to leave out a long wait at the start. They cannot go beyond the recording. `WHOLE CAPTURE` clears the limits. The same range is used by `DATA-PAGE TIME RANGE ONLY` on the Export page.

Data coverage

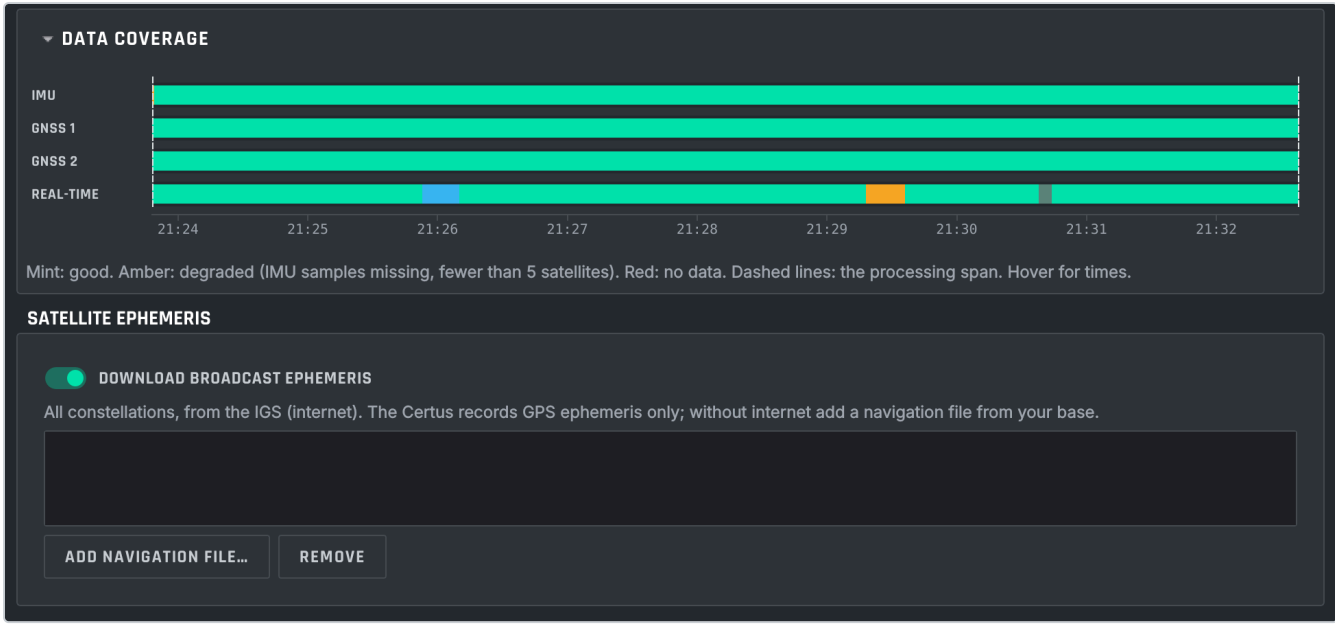


Figure 8.2 Data coverage and the ephemeris settings. Here the captures hold the Certus real-time solution only: the IMU and GNSS 1 bars are red and the real-time bar shows the two passes.

One bar per data stream over the time of the capture: **IMU**, **GNSS 1**, **GNSS 2** (when the second antenna was recorded) and **Real-time**. Mint is good, amber degraded (IMU samples missing, fewer than 5 satellites), red no data. The real-time bar takes the colour of the Certus fix type (RTK fixed mint, RTK float blue, DGNSS or SBAS amber, 2D/3D red). Dashed lines mark the processing span; hover a bar for times. The Base page shows the same bars with one more per base station.

Satellite ephemeris

- `DOWNLOAD BROADCAST EPHEMERIS` (on by default): merged broadcast ephemeris of all constellations from the IGS archives, for each day of the drive. Downloads are cached in the `pospro_cache` folder of your QGIS profile.
- The Certus records GPS ephemeris only. Without internet, add a RINEX navigation file from your base with `ADD NAVIGATION FILE…`; `REMOVE` takes it off the list. Navigation files lying next to the base observation files (same 4-character station name) are also used automatically.
- Every navigation file that is not the Certus's own or the IGS broadcast file is checked first: each orbit is compared with the Certus's orbits and, with internet, the IGS broadcast orbits of the same satellite. Orbits more than 30 m apart are left out, and the log names the file and the satellites. One bad orbit is enough to ruin a trajectory, so if

the warning comes up, take the file out of the base folder. Satellites with no reference orbit (GLONASS without internet) cannot be checked and are kept.

9 Step 2 - Installation

> WHY HERE The antenna offsets are used in every epoch of processing. They come straight from the capture, so this step is mostly a check; but a wrong or swapped antenna cannot be repaired after processing, only by processing again. `CHECK ANTENNAS` confirms them from the data itself.

Offsets are in metres from the Certus reference point, in the vehicle frame: **Forward +, Starboard +, Down +**. This is the same frame and the same values as the mapper's *Settings › Hardware*, and the page is laid out the same way. The offsets are filled from the capture when the rover files are read.

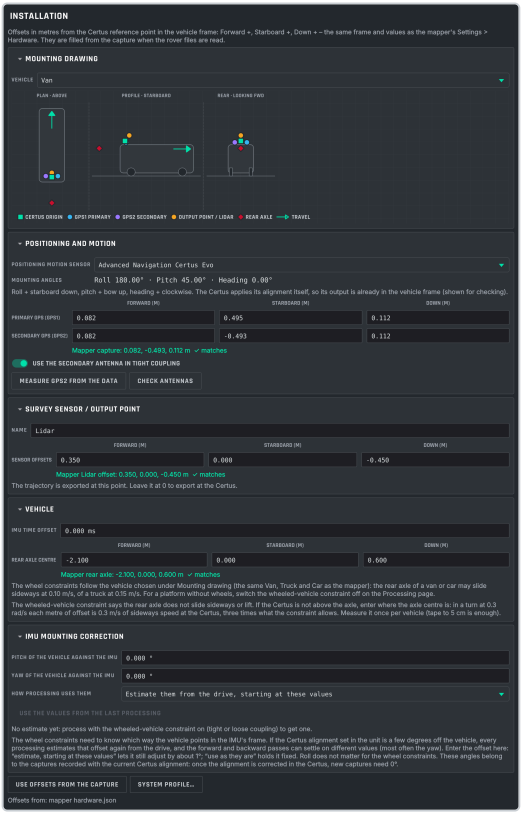


Figure 9.1 The Installation page, filled from the mapper's `hardware.json` in the capture. The green notes confirm that the values match the capture.

Mounting drawing

Choose the `VEHICLE` : Van, Truck or Car, the same vehicles and sizes as in the mapper (*Settings › Hardware*). When the rover files are read, the vehicle recorded in the capture is chosen for you. The vehicle also sets the wheel constraints used in processing (below). The drawing shows the vehicle from above, from starboard and from the rear, with the Certus origin, GPS1, GPS2, the output point (Lidar) and the rear axle as you enter them, so a sign error or a swapped axis shows at a glance.

Positioning and motion

- `POSITIONING MOTION SENSOR` : the IMU model, which sets the noise and bias values the filter uses: Advanced Navigation Certus Evo (default), Advanced Navigation Certus, Generic tactical-grade MEMS, Generic industrial MEMS.

- `MOUNTING ANGLES`: the Certus alignment (roll + starboard down, pitch + bow up, heading + clockwise) from the capture. The Certus applies its alignment itself, so its output is already in the vehicle frame; the angles are shown for checking only.
- `PRIMARY GPS (GPS1)` and `SECONDARY GPS (GPS2)`: the antenna offsets. Under each row PosPro shows what the Certus or the capture reports: green ✓ *matches* within 1 cm, amber ≠ *differs by … mm* otherwise. With the Certus dual antenna in automatic mode there is no GPS2 offset to read, and PosPro says so.
- `USE THE SECONDARY ANTENNA IN TIGHT COUPLING` (on by default): with both antennas, heading stays observable when parked or driving straight, and GNSS gaps are bridged better. It is greyed out when the captures hold no observations of the second antenna. GPS2 is used only when its offset is not zero.
- `MEASURE GPS2 FROM THE DATA`: computes the GPS1–GPS2 carrier-phase baseline from the recorded observations, turns it into the vehicle frame with the Certus attitude and adds it to GPS1. It works parked or driving, needs a few minutes with both antennas in the open, and asks before using the result (with its ± values, the antenna separation and the share of fixed epochs). Enter GPS1 first. If the antennas may be plugged in the wrong way round, run `CHECK ANTENNAS` first: swapped antennas also turn the Certus heading 180°, which would hide the swap from this measurement.
- `CHECK ANTENNAS`: see below.

Checking the antennas

If the PRIM and SEC antenna cables were plugged in the wrong way round, GPS1 (the primary receiver) is really on the secondary antenna, and every offset is wrong by the whole distance between the antennas. The Certus does not notice: its own heading simply turns 180°. `CHECK ANTENNAS` finds it from the recorded data, without a base station:

- It measures the GPS1 → GPS2 baseline from the carrier phase of both receivers (fixed epochs only), and while the vehicle drives straight above 11 km/h it puts that baseline in the vehicle frame using the **direction of travel**, which does not depend on the antennas.
- It compares the result with the offsets above: *Antennas as installed* (green), *Antennas swapped* (red: GPS2 is where GPS1 should be) or *Not conclusive* (amber: neither, or too little straight driving; parked data cannot tell, because then only the Certus heading is available).
- It also reports how far the Certus heading was from the direction of travel; about 180° confirms swapped antennas.

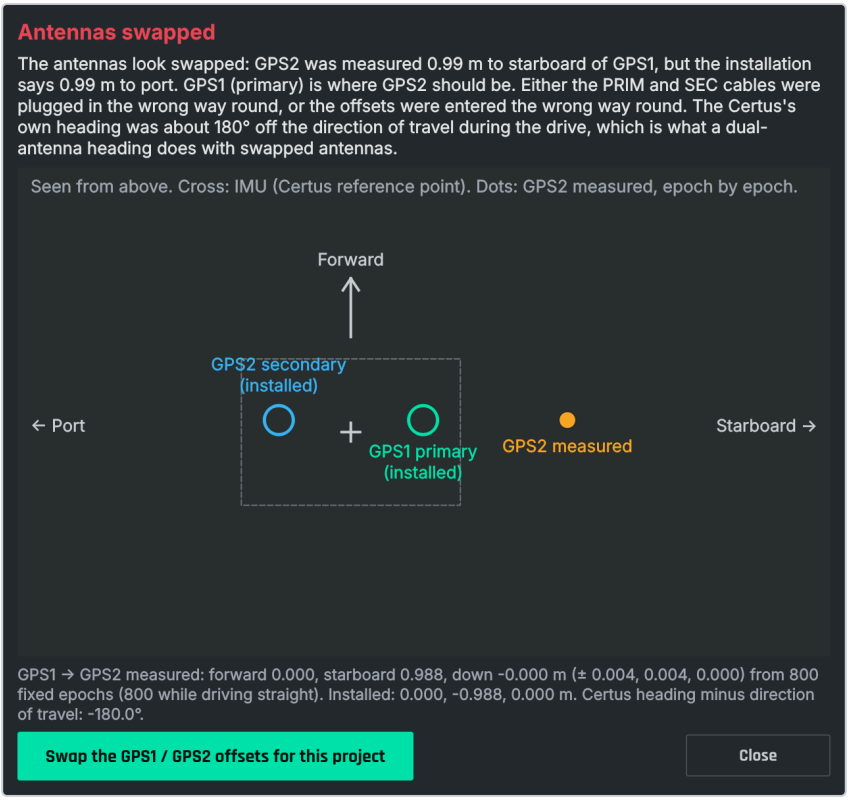


Figure 9.2 A drive recorded with the antenna cables swapped: GPS2 was measured 0.99 m to starboard of GPS1 instead of 0.99 m to port, and the Certus heading was 180° off.

When the antennas look swapped, `SWAP THE GPS1 / GPS2 OFFSETS FOR THIS PROJECT` processes the data as it was recorded: GPS1 gets GPS2's offset and the other way round, and when the Certus heading was turned 180° the INS is aligned from the direction of travel instead (Processing › Alignment *From motion*). Process again afterwards, and fix the cables on the unit for the next captures. In the Trajectory editor the measured GPS2 positions are drawn along the track as well (see *The Trajectory editor*).

Survey sensor / output point

A `NAME` (for example *Lidar*) and the `SENSOR OFFSETS` of the point you may want the trajectory exported at. It appears as a choice under `POSITION OF` on the Export page. Leave it at 0 to export at the Certus. The finalized trajectory for AlignPro and CloudPro is always at the IMU centre, whatever you enter here.

Vehicle

- **Wheel constraints** follow the vehicle chosen above: the rear axle of a van or car may slide sideways at 0.10 m/s, of a truck at 0.15 m/s. For a platform without wheels (a boat, a trolley), switch `WHEELED VEHICLE CONSTRAINT` off on the Processing page, and `ZERO VELOCITY WHEN PARKED` too if it never stands still.
- `IMU TIME OFFSET` (ms): a fixed time shift of the IMU data, for expert use; leave it at 0 for the Certus.
- `REAR AXLE CENTRE` : the wheeled-vehicle constraint says the rear axle does not slide sideways or lift. If the Certus is not above the axle, enter where the axle centre is: in a turn at 0.3 rad/s each metre of offset is 0.3 m/s of sideways speed at the Certus. Measure it once per vehicle; a tape to 5 cm is enough. The mapper's rear-axle value is read from the capture when it is there.

IMU mounting correction

The wheel constraints need to know which way the vehicle points in the IMU's frame. When the alignment set in the Certus is a few degrees off how it is really mounted, PosPro estimates the difference (pitch and yaw) during every processing and writes it to the log (*vehicle frame against the IMU: pitch … yaw …*). The forward and backward passes can settle on different values, most often in yaw. Here you set them once:

- **PITCH** / **YAW OF THE VEHICLE AGAINST THE IMU** (degrees). Roll does not matter for the wheel constraints.
- **HOW PROCESSING USES THEM**: *Estimate them from the drive, starting at these values* (default; with values entered they may still move by about 1°), or *Use these values as they are* (held fixed).
- **USE THE VALUES FROM THE LAST PROCESSING** copies the estimate of the last processing (both passes weighted by their accuracy); the line under it shows that estimate and each pass's values.

> WHICH CAPTURES The correction belongs to captures recorded with the current Certus alignment. Once the alignment is corrected in the Certus, its output is already in the vehicle frame: new captures need 0°.

USE OFFSETS FROM THE CAPTURE (shown when the capture holds offsets) sets GPS1, GPS2, the Lidar offset, the rear axle and the vehicle from the mapper's `hardware.json`, or else from the Certus installation record (packet 185) and dual-antenna set-up (packet 196), and lists what it set. The line under it says where the offsets came from (*Offsets from: mapper hardware.json*, or *Offsets entered by hand*). The Installation step turns amber while the GPS1 offset is zero.

SYSTEM PROFILES

A system profile keeps the checked offsets of one mapper system – the GPS1 and GPS2 antenna offsets, the rear axle, the IMU model and the vehicle (AlignPro adds the Lidar lever arm, mounting angles and latency to the same profile) – so once you have checked them the next job from the same system starts right. Profiles are kept on this computer; **EXPORT…** and **IMPORT…** move them to another.

- **Save:** check the offsets first (measure them, run **CHECK ANTENNAS**), then press **SYSTEM PROFILE…** on the Installation page. **NEW…** makes a profile for this job's system with this job's values; **SAVE THIS JOB'S VALUES TO THE PROFILE** updates the selected profile (only the checked sections; the Lidar section AlignPro saved is kept). The result of the last antenna check is noted with the antenna offsets.
- **Which system:** reading the rover files records the mapper unit's serial number (from a survey line's `session.json`) and the Certus serial number (from `hardware.json`). A profile matches on the unit first, then on the Certus.
- **Use:** after the rover files are read, when a profile matches and its values differ from the job's, PosPro lists the differences and asks. **USE PROFILE VALUES** fills the Installation page (the line under it then reads *Offsets from: system profile <name>*) and updates the shared antenna offsets; **KEEP THIS JOB'S VALUES** leaves it. Each version of a profile is offered once per job.
- Why it matters: the capture's `hardware.json` holds whatever was saved on the unit when it recorded, and cannot show swapped antenna cables. The profile holds what you have checked.

10 Step 3 - Base stations

> **WHY HERE** Every PPK position is relative to the base, so the base's own position decides where the whole trajectory sits. It is computed (PPP or CSRS-PPP) before processing; the RINEX header position is only approximate.

Differential processing (PPK) and tight coupling need one base station that covers the whole drive. Every point of the trajectory inherits the base position's error and its frame, so give the base a precise position before processing. The Base page is not needed for PPP or for the real-time mode (the step then reads *not needed (PPP)* or stays grey).

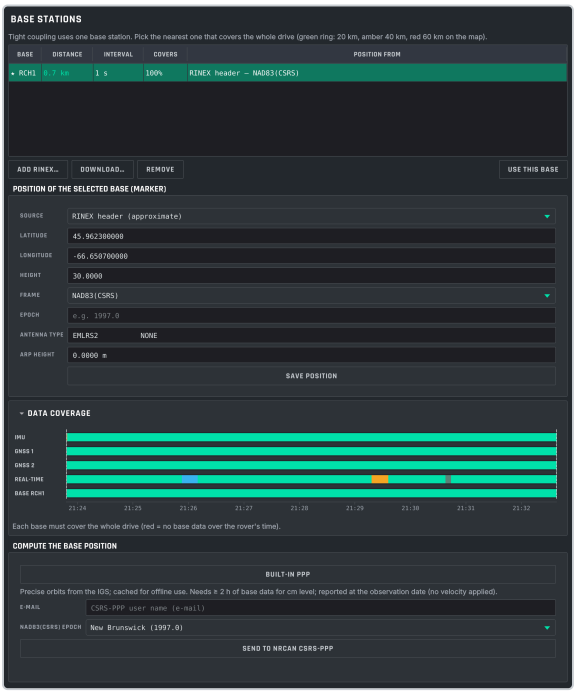


Figure 10.1 The Base page with a Reach base station's RINEX added: 0.7 km from the drive, 1 s, covering the whole drive. Its position is still the approximate one from the RINEX header (amber): compute it with built-in PPP or CSRS-PPP before processing.

The base table

COLUMN	SHOWS
Base	The station name. ★ marks the base used for processing.
Distance	From the drive: green up to 20 km, amber up to 40 km, red beyond.
Interval	The observation interval of the file.
Covers	The share of the rover's time that the base file covers. It should be 100 %.
Position from	RINEX header, entered, built-in PPP or CSRS-PPP, with the frame and epoch.

- **ADD RINEX…** adds a base from RINEX observation files (`.obs .rnx .crx .??o .??d .gz`). Select several files of the same station to add them as one base; their observations are merged (they must list the same observation types, otherwise add them as separate bases). Compact (Hatanaka) RINEX is decompressed automatically.
- **DOWNLOAD…** opens the base-station finder (below).
- **REMOVE** removes the selected base; **USE THIS BASE** makes it the one used for processing (the first base added is used until you choose another).

The bases also appear on the QGIS map, in the group *PosPro: <project>*: the base used as a large red triangle, the others grey, with distance rings at 20, 40 and 60 km.

Position of the selected base (marker)

The coordinates are those of the marker. Choose the **SOURCE** (RINEX header (approximate), Entered / finalized, Built-in PPP, NRCan CSRS-PPP), enter **LATITUDE** and **LONGITUDE** in decimal degrees and the ellipsoidal **HEIGHT**, the **FRAME** (NAD83(CSRS), ITRF2020 / IGS20, WGS 84, NAD83(2011)) and its **EPOCH**, the **ANTENNA TYPE** (from the RINEX header; used with the IGS antenna calibrations) and the **ARP HEIGHT**, then press **SAVE POSITION**. The frame and epoch of the base become those of the project: the trajectory and every export are in that frame.

Compute the base position

BUTTON	USE WHEN	RESULT
BUILT-IN PPP	No CSRS-PPP account, or offline once the products are cached	Static PPP with downloaded precise orbits and clocks (ESA, IGS or CODE; finals first, then rapids) and the IGS antenna file. ITRF2020 at the observation date, transformed to NAD83(CSRS) at the same date when the base frame is NAD83(CSRS); no velocity applied. Needs at least 2 hours of base data for centimetre level; PosPro warns when the file is shorter.
SEND TO NRCAN CSRS-PPP	Internet and an NRCan account	Enter the **E-MAIL** you use as your NRCan user name and choose the **NAD83(CSRS) EPOCH**: New Brunswick (1997.0), Federal (2010.0) or Observation date. PosPro sends the file, waits for the result and enters the NAD83(CSRS) position at that epoch.

The result of each computation is listed under the buttons (built-in PPP: date, hours of data, products and σ). For work tied to NAD83(CSRS) epoch 1997.0, use CSRS-PPP with that epoch: the built-in PPP gives the position at the observation date, which differs by the crustal motion since 1997.

> APPROXIMATE BASE POSITIONS The Base step turns amber while the base position comes from the RINEX header, and **PROCESS** asks before using it: *The base position comes from the RINEX header and may be metres off; every point of the trajectory inherits that error.*

The base-station finder

DOWNLOAD… opens *PosPro – base stations near the drive*: a map of the reference stations within 150 km of the drive, with the drive itself in red and dashed rings at 20, 40 and 60 km. Read the rover files first; the map centres on the drive. The finder needs the internet.

NETWORK	SYMBOL	DOWNLOAD FROM POSPRO
NRCan CACS, NS active control, NRCan regional	squares (red, pink, dark red)	Yes, free
IGS	mint star	Yes, free
NOAA CORS	blue triangle	Yes, free
Can-Net, SmartNet, BrandtNet, TopNET, GEODNET, Other RTK	diamond, pentagon, hexagon, triangle, circle, cross	Through the network's portal (**OPEN PORTAL**, then **IMPORT FILE…**), or with your account where the network allows it

1. Untick networks in the legend under the map to hide them. Large symbols are the three nearest stations of each network; the table lists them with **NETWORK**, **STATION**, **KM**, **RATES** and **ACCESS** (Free download, Your

account, Portal only).

2. Click a station on the map or a row. The card shows its name, operator, distance (green within 20 km, amber within 40, red beyond), position, receiver and antenna, status (for the commercial networks, NRCan's check of their tie to NAD83(CSRS)), rates and access, and the days of the drive.
3. Choose the **RATE** (rates under 5 s come as 15-minute files, others as daily files), leave **PRECISE ORBITS, CLOCKS AND ANTENNA FILE** on, and press **DOWNLOAD**. A progress card shows the file, the megabytes and the time, with **CANCEL**. Networks that need your account ask for it once and keep it in the QGIS password store.
4. When the download finishes the finder closes and the station is added to the base table, with a message saying where the files were saved (the project's `work/base` folder).

Station lists are cached for a week; **REFRESH STATION LISTS** loads them again. The commercial stations come from NRCan's list of networks tied to NAD83(CSRS), so they cover Canada only.

11 Step 4 - Processing

> **WHY HERE** With the data read, the antennas checked and the base positioned, processing has everything it needs. The defaults (tight coupling, both antennas, both passes) are the most accurate and suit almost every drive.

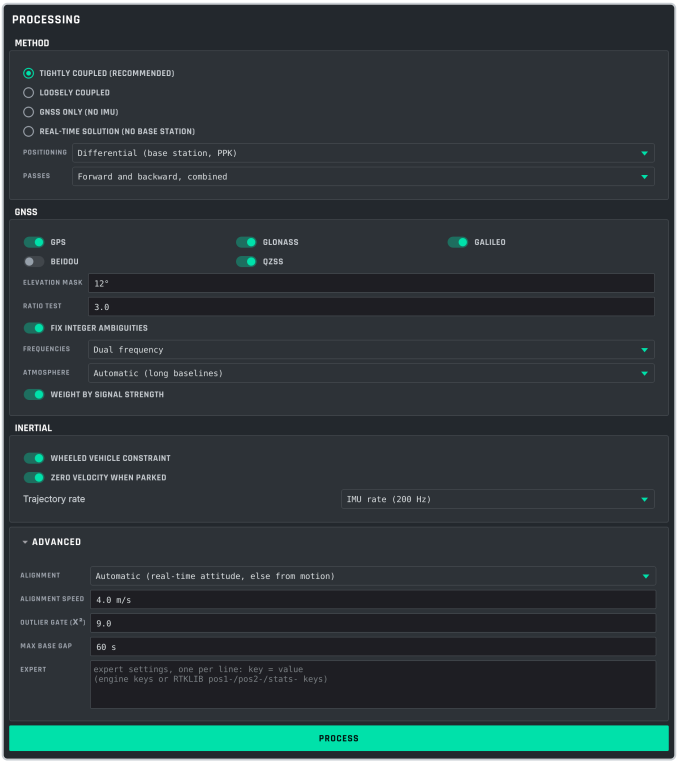


Figure 11.1 The Processing page with its default settings (tightly coupled, PPK, both passes) and the **ADVANCED** section opened.

Method

METHOD	WHAT IT DOES	NEEDS
TIGHTLY COUPLED (RECOMMENDED)	Raw carrier phase and code from rover (both antennas) and base go straight into the inertial filter. Keeps fixing with fewer than 4 satellites and repairs cycle slips with the IMU.	Raw IMU, raw GNSS, a base station with a position
LOOSELY COUPLED	GNSS positions are computed first (RTKLIB) and then blended with the IMU.	Raw IMU, raw GNSS; a base for PPK, or PPP without a base
GNSS ONLY (NO IMU)	The RTKLIB kinematic solution, forward and backward, combined. No attitude: roll and pitch are 0 and the heading follows the direction of travel. For comparison, or when there is no IMU data.	Raw GNSS; a base for PPK, or PPP
REAL-TIME SOLUTION (NO BASE STATION)	The Certus's own real-time trajectory (packet 20), cleaned in Trajectory Review. For when there is no base station, or RTK was used in the field.	The real-time solution in the captures

- **POSITIONING**: *Differential (base station, PPK)* or *Precise point positioning (no base)*. PPP works with loose coupling and GNSS only; tight coupling always needs a base. With PPP the trajectory is in ITRF2020 at the observation date, not NAD83(CSRS), and PosPro sets the project frame that way and warns.
- **PASSES**: *Forward and backward, combined* (the normal case: the two passes are combined by their accuracy, and their difference is the separation shown on the Quality page) or *Forward only (quick check)*.

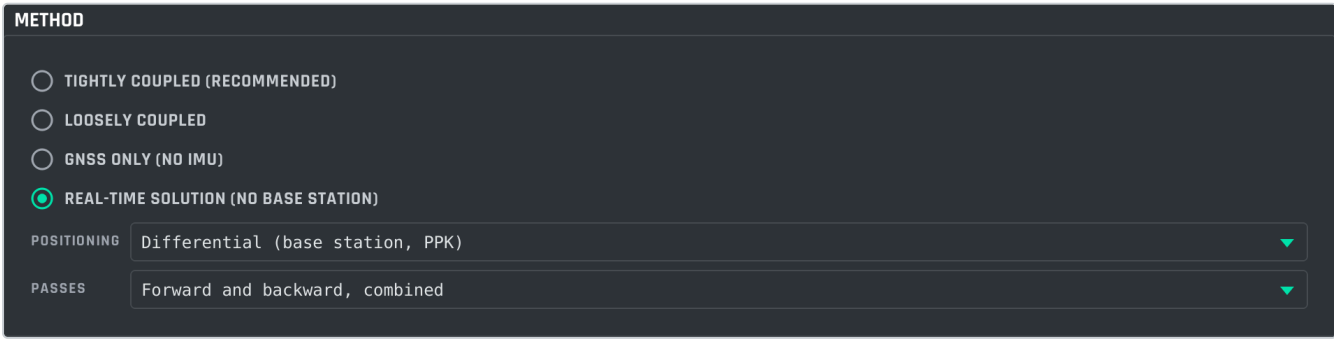


Figure 11.2 The real-time mode selected. It uses the Certus solution as recorded; the positioning and passes settings do not apply to it.

GNSS

- Constellations: **GPS**, **GLONASS**, **GALILEO**, **BEIDOU**, **QZSS** (GPS, GLONASS, Galileo and QZSS by default).
- **ELEVATION MASK** (12° by default): satellites lower than this are not used.
- **RATIO TEST** (3.0): how clearly the best integer ambiguity set must win before it is accepted as fixed. Higher is safer but fixes less.
- **FIX INTEGER AMBIGUITIES**: off gives a float solution only.
- **FREQUENCIES**: *Dual frequency* or *L1 only*.
- **ATMOSPHERE**: estimates the difference in ionosphere (per satellite) and troposphere between rover and base. *Automatic (long baselines)* turns it on beyond 10 km or 150 m height difference, where it keeps ambiguities fixable; on short baselines it only adds noise. *Always estimate*, *Off (short baselines)*.
- **WEIGHT BY SIGNAL STRENGTH**: weak signals (trees, buildings, multipath) count less than strong ones.

Inertial

- **WHEELED VEHICLE CONSTRAINT**: the vehicle does not slide sideways or jump; keeps the solution tight in GNSS gaps. How much sideways slip it allows comes from the vehicle on the Installation page (van and car 0.10 m/s, truck 0.15 m/s). Switch it off for a platform without wheels.
- **ZERO VELOCITY WHEN PARKED**: zero-velocity updates while the vehicle stands still.
- **TRAJECTORY RATE**: *IMU rate (200 Hz)*, 100, 50, 20 or 10 Hz.

Advanced

- **ALIGNMENT**: how the filter starts its attitude: *Automatic (real-time attitude, else from motion)*, *Real-time attitude from packet 20*, or *From motion (GNSS course, drive straight)*.
- **ALIGNMENT SPEED** (4.0 m/s): the speed needed to align from motion.
- **OUTLIER GATE (X²)** (9.0): GNSS measurements that disagree more than this with the filter are rejected.
- **MAX BASE GAP** (60 s): the longest interval between base observations that tight coupling bridges.
- **EXPERT**: extra settings, one per line as `key = value`. Keys starting with `pos`, `stats`, `misc` or `ant` go to RTKLIB; all others go to the inertial engine. Lines starting with `#` are ignored.

Process

Press **PROCESS**. PosPro first checks that the rover files are read and, for PPK and tight coupling, that a base is chosen (and asks when its position is only from the RINEX header). Then, in the background:

1. the rover files are read again if they changed;
2. for loose coupling, GNSS only and PPP: the GNSS solutions, forward and backward;
3. the inertial passes, forward and backward at the same time (for the real-time mode: the Certus solution as one pass);
4. the passes are combined; your Trajectory Review decisions (poor, interpolate) are applied again;
5. the trajectory is finalized for AlignPro and CloudPro.

The status line shows the stage and the progress bar the progress; **STOP** ends the run. The project box turns red and PosPro in the chain turns blue (*running*); another computer cannot process the same stage at the same time (*… is being processed by … on … since …*). When the run finishes, the Quality page opens. A run that fails or is stopped keeps the earlier results, and PosPro in the chain turns red (*failed*).

12 Step 5 - Quality

> WHY HERE Judge the result before building on it. The share of fixed ambiguities, the forward/backward separation and the longest IMU-only gap say whether to go on, change a setting, or fix an input and process again.

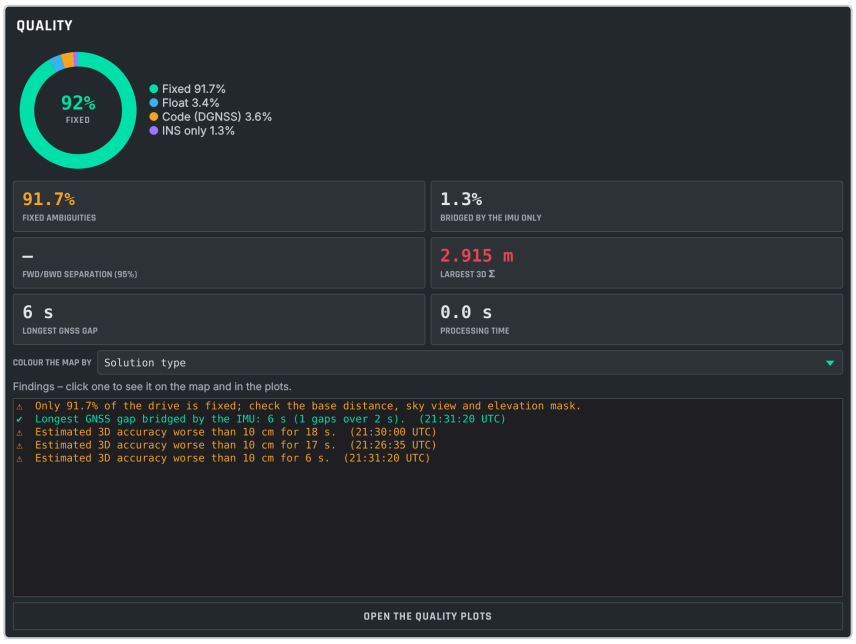


Figure 12.1 The Quality page for a real-time trajectory. With one pass there is no forward/backward separation; see the note below.

The donut shows the share of each solution type (Fixed mint, Float blue, PPP light blue, Code (DGNSS) amber, Single red, INS only violet). The cards:

CARD	GREEN	AMBER	RED
fixed ambiguities	≥ 95 %	≥ 75 %	below
bridged by the IMU only	share of the time without GNSS updates		
fwd/bwd separation (95%)	< 5 cm	< 20 cm	above
largest 3D σ	< 10 cm	< 50 cm	above
longest GNSS gap	the longest stretch bridged by the IMU only		
processing time	of the last run		

> SINGLE-PASS RESULTS With *Forward only* and in the real-time mode there is no backward pass to compare with, so the separation card shows –.

A good mobile-mapping trajectory is over 95 % fixed, has a forward/backward separation under 5 cm (95 %), and no 3D σ worse than 10 cm outside GNSS gaps. The Quality step is green from 95 % fixed, amber below.

Findings

A plain-language list: the share fixed, the longest GNSS gap bridged by the IMU and the number of gaps over 2 s, the forward/backward separation (95 % and maximum), up to five stretches where the passes disagree by more than

10 cm, up to three stretches where the estimated 3D accuracy is worse than 10 cm, and the IMU biases estimated at the end. Findings with a time show it in UTC; click one to mark that place on the map and zoom the quality plots to it (the plots open if they are closed).

The map

PosPro puts its layers in the group *PosPro: <project>* of the QGIS layer panel: **Real-time track** (what the Certus computed in the field, shown as soon as the files are read), **Base stations** and **Base distance rings**, **Trajectory** (one point per second, with the time, solution, satellites, height, attitude, speed, σ, separation and gap as attributes), **Trajectory line** and, after Trajectory Review, **Review windows**. `COLOUR THE MAP BY` colours the trajectory points by *Solution type*, *Estimated accuracy (3D σ)* or *Forward/backward separation* (classes < 2 cm, 2–5, 5–10, 10–30, 30–60, > 60 cm).

The quality plots

`OPEN THE QUALITY PLOTS` opens *PosPro – quality plots* in a window of its own (the same standard window as PosPro: a window of its own).

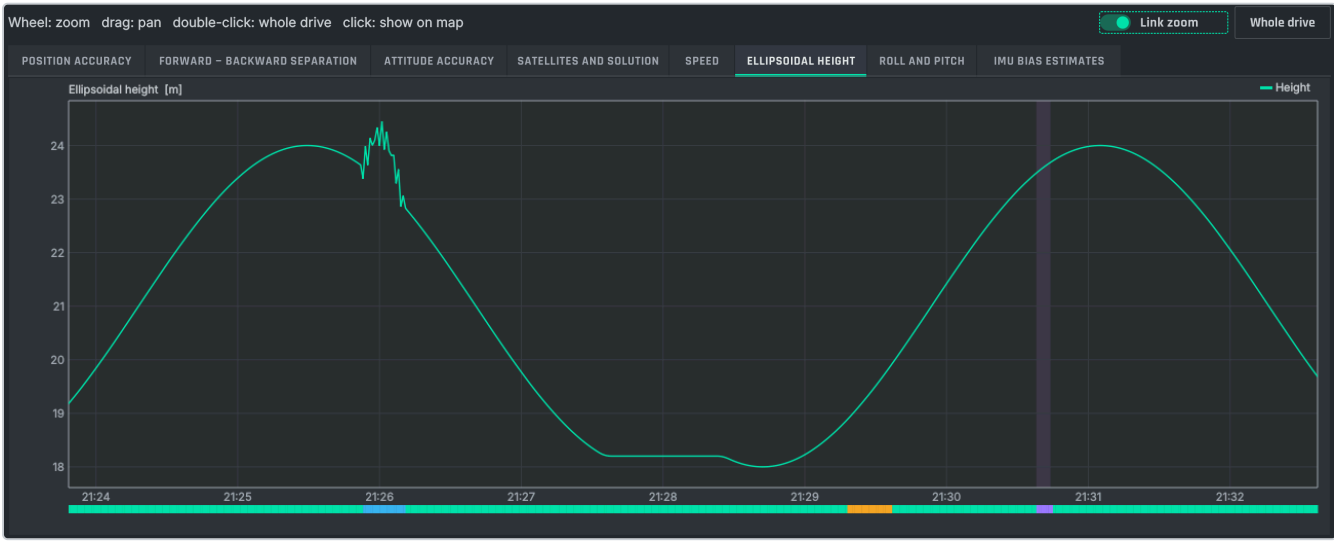


Figure 12.2 The quality plots, Ellipsoidal height tab: the two passes of the drive with the gap between them. The strip under the plot shows the solution type.

The tabs: **Position accuracy** (σ north, east, down), **Forward – backward separation** (north, east, down, where both passes have a solution), **Attitude accuracy** (σ roll, pitch, heading), **Satellites and solution** (satellites and the ambiguity ratio ÷ 10), **Speed**, **Ellipsoidal height**, **Roll and pitch** and **IMU bias estimates** (accelerometers in mg, gyros in deg/h).

Wheel to zoom, drag to pan, double-click (or `WHOLE DRIVE`) for the whole drive. Hover for the values at a time. A click marks the time with an orange line and shows that place on the map with an orange circle. With `LINK ZOOM` on, all tabs follow the same time range. Under every plot a strip shows the solution type in the map colours, and GNSS gaps (INS only) are shaded violet.

13 Step 6 - Calibration

> WHY HERE Solving the lever arms needs a processed drive and means processing again with the result. Doing it before any review means no hand edits are lost. It is optional: for a known installation, skip it.

The lever-arm solver estimates the IMU-to-antenna offsets from the drive itself and shows them beside the ones you entered, so a mis-measured antenna shows up before it spoils a survey. It runs its own tightly coupled processing of the drive (it needs the rover files read, a base station and *Tightly coupled*); measure the offsets as well as you can first, and let the solver check and refine them.

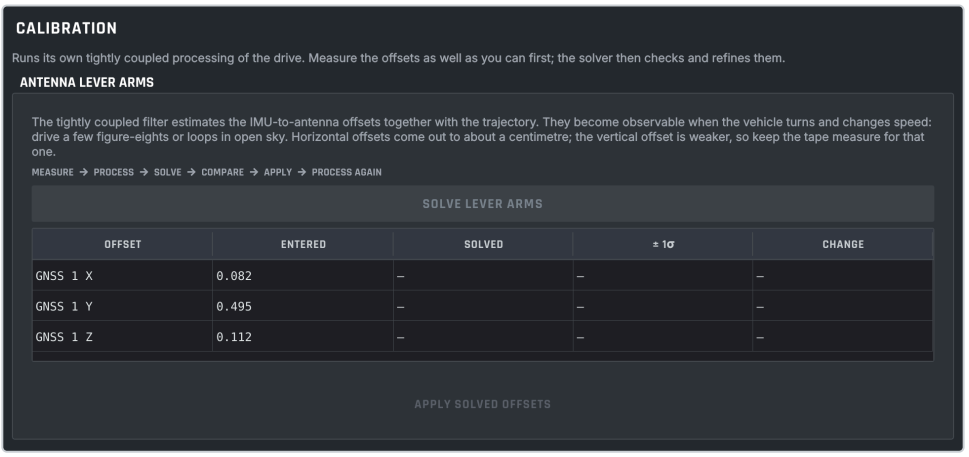


Figure 13.1 The Calibration page before solving: the entered GPS1 offsets, nothing solved yet.

1. Process the drive with **TIGHTLY COUPLED** and a base station, fixed for most of the drive. The solver needs tight coupling; in any other mode it says so (Figure 13.2).
2. Press **SOLVE LEVER ARMS**. Both passes run again with the offsets as unknowns, about as long as a normal processing run, and are combined by their accuracy.
3. Read the table: **OFFSET** (GNSS 1 and, when used, GNSS 2; X forward, Y right, Z down), **ENTERED**, **SOLVED**, **± 1Σ** and **CHANGE**. A change in amber is larger than three standard deviations and 2 cm.
4. Re-measure an amber offset if you can. Otherwise press **APPLY SOLVED OFFSETS**: the solved values go to the Installation page. Process again so the trajectory uses them, and enter the same values in the Certus and the mapper Settings if you keep them.

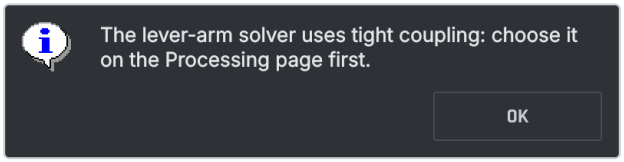


Figure 13.2 The solver needs tight coupling.

The offsets become observable when the vehicle turns and changes speed: drive a few figure-eights or loops in open sky. Horizontal offsets come out to about a centimetre; the vertical offset is weaker, so keep the tape measure for that one. A straight drive cannot separate the forward offset from the IMU's own errors.

14 Step 7 - Trajectory Review

> WHY HERE Hand edits belong to one processed trajectory (processing again takes hand selections off the list), so they come once the processing is final. Review is for what the data cannot give, such as a wrong fix under trees; wrong inputs are fixed at their own step and processed again.

Trajectory Review finds where the trajectory misses your limits, lets you look at each place in the Trajectory editor, on the map and in the plots, and records what you decide for each. Decisions are kept with the project and carried to AlignPro and CloudPro.

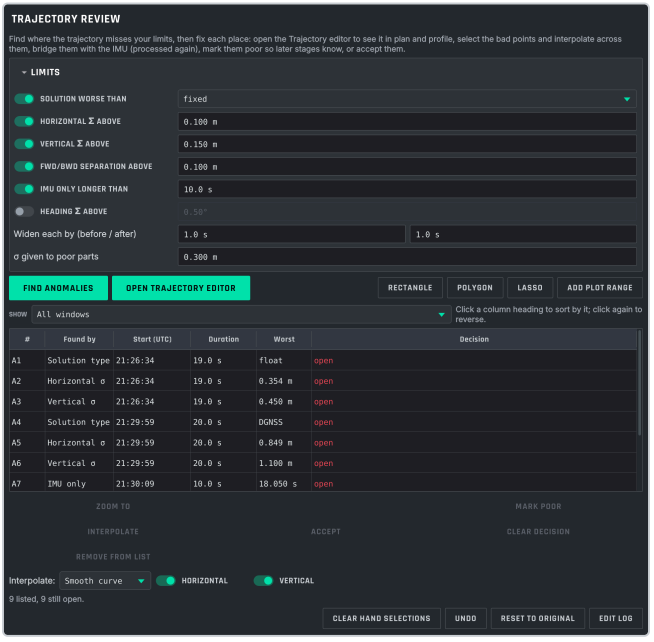


Figure 14.1 Anomalies found with the default limits on the simulated drive: the float spell under trees, the DGNSS spell with a sideways jump and the short GNSS gap. All are still open.

Limits

LIMIT	DEFAULT	FINDS
SOLUTION WORSE THAN	fixed (on)	Samples worse than fixed, float, DGNSS or single (IMU-only samples are left to the IMU limit)
HORIZONTAL Σ ABOVE	0.100 m (on)	Horizontal accuracy worse than the limit
VERTICAL Σ ABOVE	0.150 m (on)	Vertical accuracy worse than the limit
FWD/BWD SEPARATION ABOVE	0.100 m (on)	The passes disagreeing horizontally by more than the limit
IMU ONLY LONGER THAN	10.0 s (on)	GNSS gaps longer than the limit
HEADING Σ ABOVE	0.50° (off)	Heading accuracy worse than the limit
WIDEN EACH BY (BEFORE / AFTER)	1.0 s / 1.0 s	A margin added to each window, because a drop in quality is often reported a little late

Σ GIVEN TO POOR PARTS	0.300 m	The σ that **MARK POOR** gives

Flagged samples less than 1.5 s apart join into one window.

Finding and selecting

- **FIND ANOMALIES** checks the trajectory as processed (before any review edit) against the limits and lists the windows; a message says how many were found. Run it again after changing the limits.
- **RECTANGLE**, **POLYGON**, **LASSO**: select part of the trajectory on the QGIS map. Rectangle: drag. Polygon: click the corners; right-click or double-click to finish. Lasso: drag around the part. Esc cancels. The parts of the trajectory inside the shape are added to the list as *Selected by hand*: parts less than 30 s apart become one window, and a drawing that catches more than 20 separate parts becomes one window from the first to the last. The same window drawn twice is listed once.
- **ADD PLOT RANGE**: adds the time range the quality plots are zoomed to. Open the plots and zoom first.

Processing finds the anomalies by itself with the limits above, so the list is ready when processing ends. Hand selections belong to the trajectory they were drawn on: processing again takes them off the list (decisions already made on them stay). **REMOVE FROM LIST** takes the selected **S** rows off, and **CLEAR HAND SELECTIONS** takes them all off.

SHOW narrows the list to the open windows, the ones found by the limits (A), the hand selections (S) or the decided ones; click a column heading to sort by it (again to reverse) – for example by *Worst* to look at the largest first. To get rid of old hand selections: **SHOW** › *Selected by hand*, select them (Ctrl+A) and **REMOVE FROM LIST**, or **CLEAR HAND SELECTIONS**.

The list shows the window (**A** found, **S** selected by hand), what found it, the start (UTC), the duration, the worst value and the decision (*open* in red until you decide). The windows are drawn on the map in the layer *Review windows*: red open, blue bridged with the IMU, amber poor, violet interpolated, mint accepted. Double-click a row, or **ZOOM TO**, to see it on the map and in the plots.

Recording holes

When the recording lost INS data (mapper 2.9.0 recordings lost about a third, in holes of 0.5–3 s), the trajectory has no records in those holes and every program draws a straight line across them. The finalized trajectory lists them (`trajectory.json`, `recording_holes`) and Trajectory Review says at the top how much is missing, e.g. *Recording holes: the trajectory has no data for 35% of its time (841 holes, longest 110.4 s).* Bridging them one by one here does not bring the motion back; **StripPro** recovers it from the Lidar.

Problem areas from the point cloud

Some trajectory errors are invisible to PosPro's own checks, for example when both passes hold the same wrong fix: the limits above see nothing, but the points of that line do not lie on the other lines' points. CloudPro measures this (*project menu › Check line separation* in CloudPro) and lists each local problem area with the line at fault and the time its points were taken. PosPro reads that list from CloudPro's folder of the same job and shows it here:

- In the review list as **C** rows, found by *Point cloud*, widened by the same margins as the other windows. *Worst* gives the largest separation and the line, e.g. *17 cm, Line 3*. When CloudPro could not tell which of two lines is at fault, each line gets its own row (*C3a*, *C3b*) with the other line in brackets. **SHOW** › *Found in the point cloud (C)* lists only them.
- In the Trajectory editor as a pink stretch with its label on the plan, and a pink band in the profile (Figure 14.2).

From CloudPro, **OPEN IN POSPRO** in the LINE SEPARATION tab opens this page on the same job with **SHOW › FOUND IN THE POINT CLOUD (C)** and the chosen area selected.

Decide on them like any other window: bridge with the IMU, mark poor, interpolate or accept. They are not stored in PosPro: they come from CloudPro's latest check. Once PosPro finalizes another trajectory, they are marked *earlier trajectory* (they no longer count as open in the Review step); CloudPro updates its points, and its next check shows whether the problem has gone.

The Trajectory editor

OPEN TRAJECTORY EDITOR (next to **FIND ANOMALIES**) opens the processed trajectory in its own window: the **plan view** on top, the **height profile** (height against time) below, and on the right the **sections** of the job and the **review list**. Everything you decide here is the same decision as on the Review page, so both always agree.

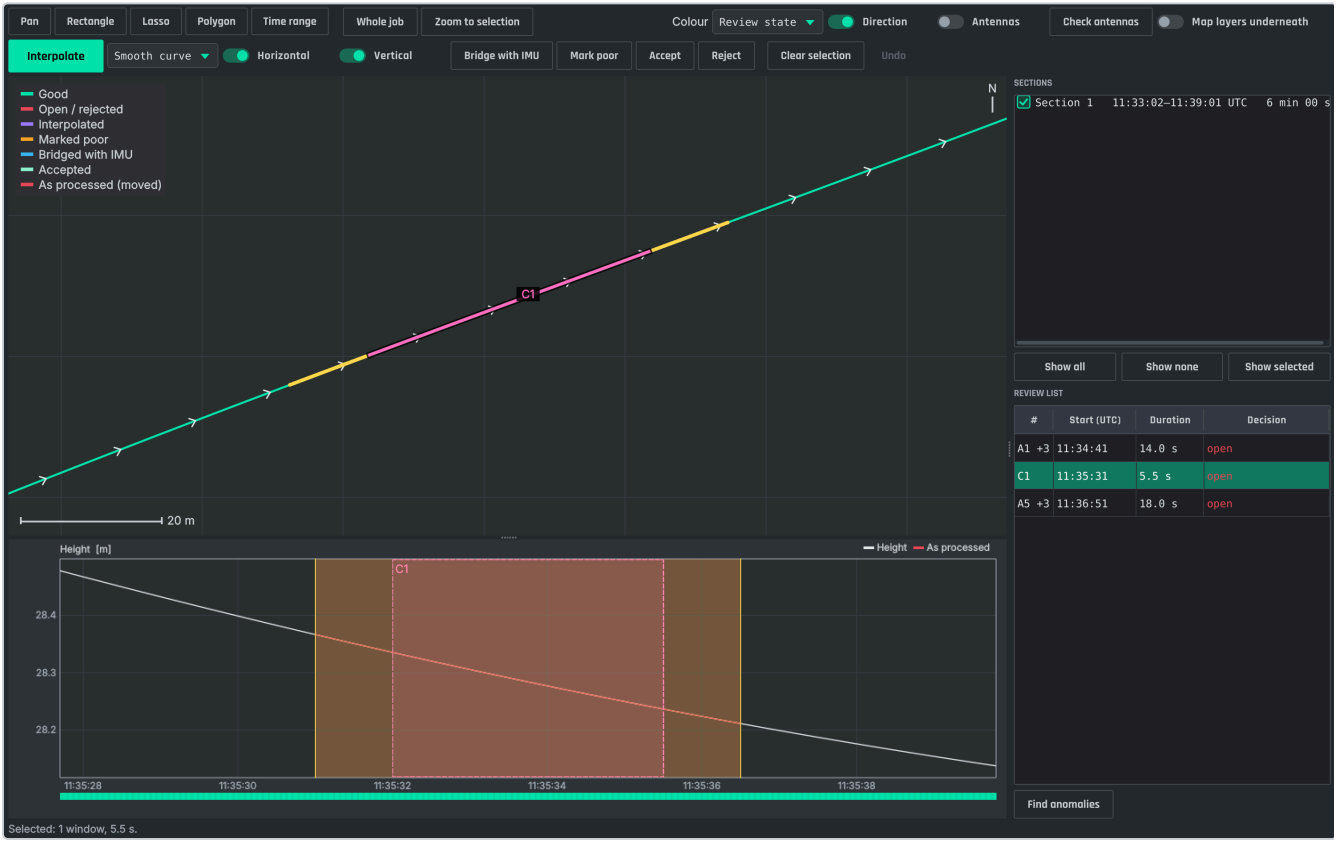


Figure 14.2 A problem area found in the point cloud (C1, pink) chosen in the review list: selected in yellow, with the margins either side.

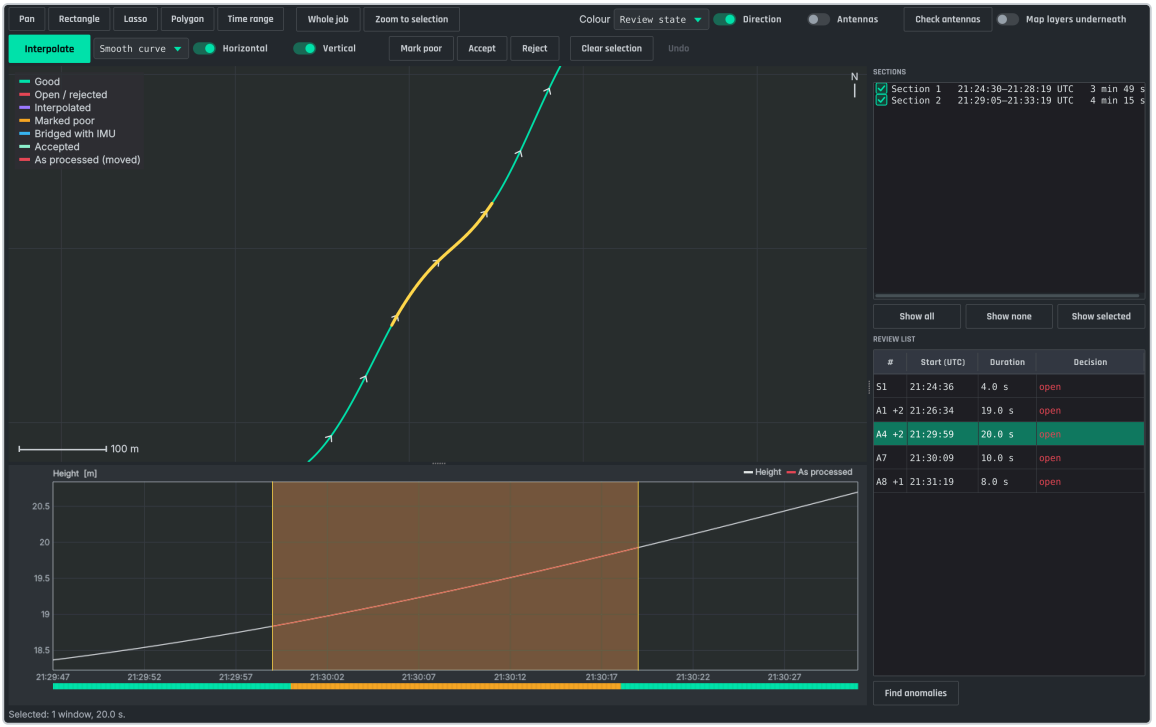


Figure 14.3 The Trajectory editor on an anomaly: the window chosen in the review list is selected (yellow) in the plan and in the profile, where the height as processed (red) jumps about. The band under the profile shows the solution type (mint fixed, blue float).

The views are linked. Move the mouse over the plan and the profile shows a line at that moment with its height; move it over the profile and a ring marks the same place in the plan. Click a point in either and it is marked in both. Zoom or pan the plan (wheel, drag) and the profile shows the time of what is in view; zoom the profile and the plan follows when that stretch is off screen. The line under the views gives the time, position, height, solution type and review state under the mouse.

TOOL	USE
PAN	Drag to move, wheel to zoom, click a point to mark it in both views, double-click to see the whole job. The middle mouse button pans with any tool.
RECTANGLE, LASSO, POLYGON	Select bad points in the plan view (polygon: click the corners, right-click or double-click to finish; Esc cancels). The selection is shown in both views; parts less than 3 s apart become one window.
TIME RANGE	Drag across the profile to select a time range, e.g. a height spike.
WHOLE JOB, ZOOM TO SELECTION	Fit both views to the shown sections or to the selection.
COLOUR	*Review state*: mint good, red open or rejected, violet interpolated, amber poor, blue bridged with the IMU, pale mint accepted. *Solution type*: the mapper's solution colours. Positions the edits moved are also drawn as they were processed, in faint red.
DIRECTION	Chevrons along the track pointing the way the vehicle drove (on by default).
ANTENNAS, CHECK ANTENNAS	ANTENNAS draws GPS1 (mint) and GPS2 (blue) along the track from the Installation offsets, and at the point under the mouse or clicked, the vehicle: the IMU joined to both antennas. After CHECK ANTENNAS (see *Installation › Checking the antennas*) the GPS2 positions it measured are drawn as orange dots: on the blue line when all is well, on the other side of GPS1 when the antennas are swapped.
MAP LAYERS UNDERNEATH	Draws the layers of the QGIS map (your basemap, roads) under the plan view.

Sections	The job split where the vehicle stood still for 30 s or more, or the data stops. Untick sections to hide them; **SHOW SELECTED** shows only the highlighted ones; double-click a section to zoom to it.
Review list	One row per place (a place found by several limits shows as e.g. *A1 +3*). Click a row to select its window in both views and zoom there.

With a selection, choose what to do with it (the buttons above the views):

BUTTON	EFFECT
INTERPOLATE	Replaces the selected positions between the good points on each side, **HORIZONTAL** and/or **VERTICAL**. *Smooth curve* leaves and meets the good points at their own speed and direction, so it follows a bend; *Straight line* joins them directly (good for a height spike on level ground). A selection at the very start or end of the trajectory is carried on from its one good side at that point's speed and direction.
BRIDGE WITH IMU	The same as on the Review page: the IMU carries the trajectory there after **PROCESS AGAIN** (shown when waiting). Tightly and loosely coupled only.
MARK POOR, **ACCEPT**	As on the Review page.
REJECT	Marks it bad, to fix later: it stays *open* in red.
UNDO	Takes the last decision back.

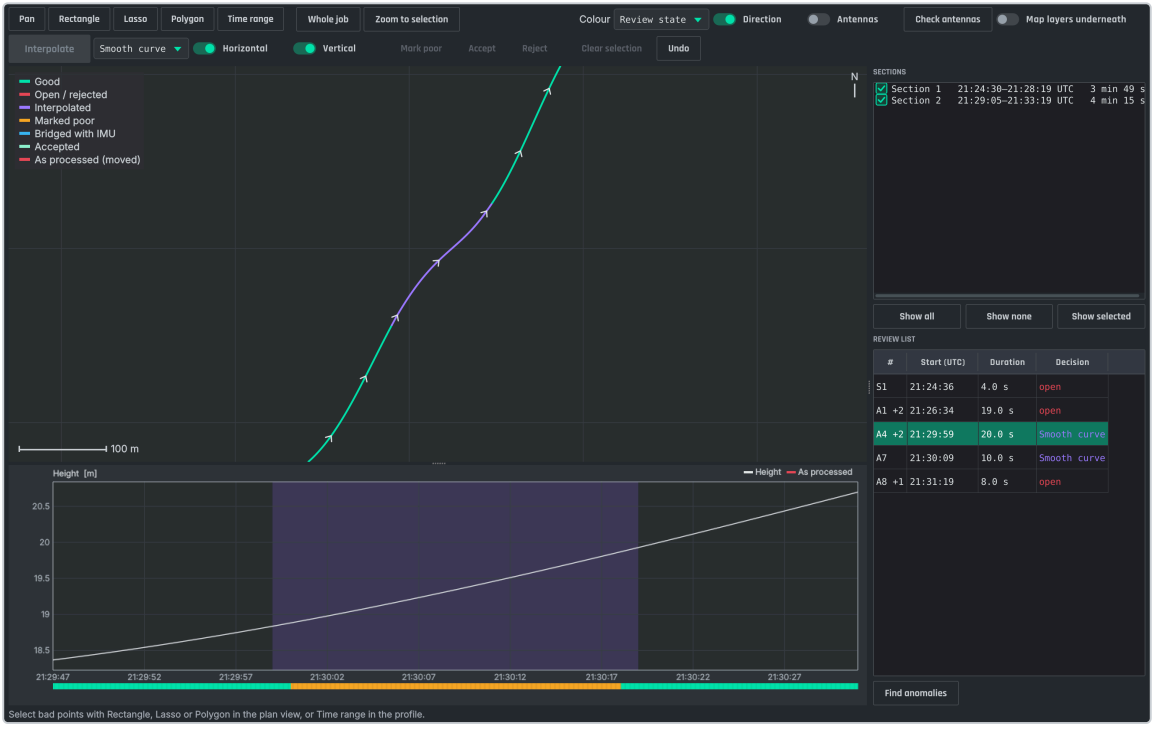


Figure 14.4 A sideways jump fixed with a smooth curve, horizontally only: the trajectory now (violet) follows the bend, and the positions as processed are drawn in faint red beside it.

> INTERPOLATE OR BRIDGE WITH THE IMU? Interpolation is best for short stretches: spikes, a few seconds of wrong fixes. Over a longer stretch the IMU knows the turns and bumps that a curve cannot guess, so **BRIDGE WITH IMU** usually follows the road better; PosPro reminds you when you interpolate more than 10 s of a coupled trajectory.

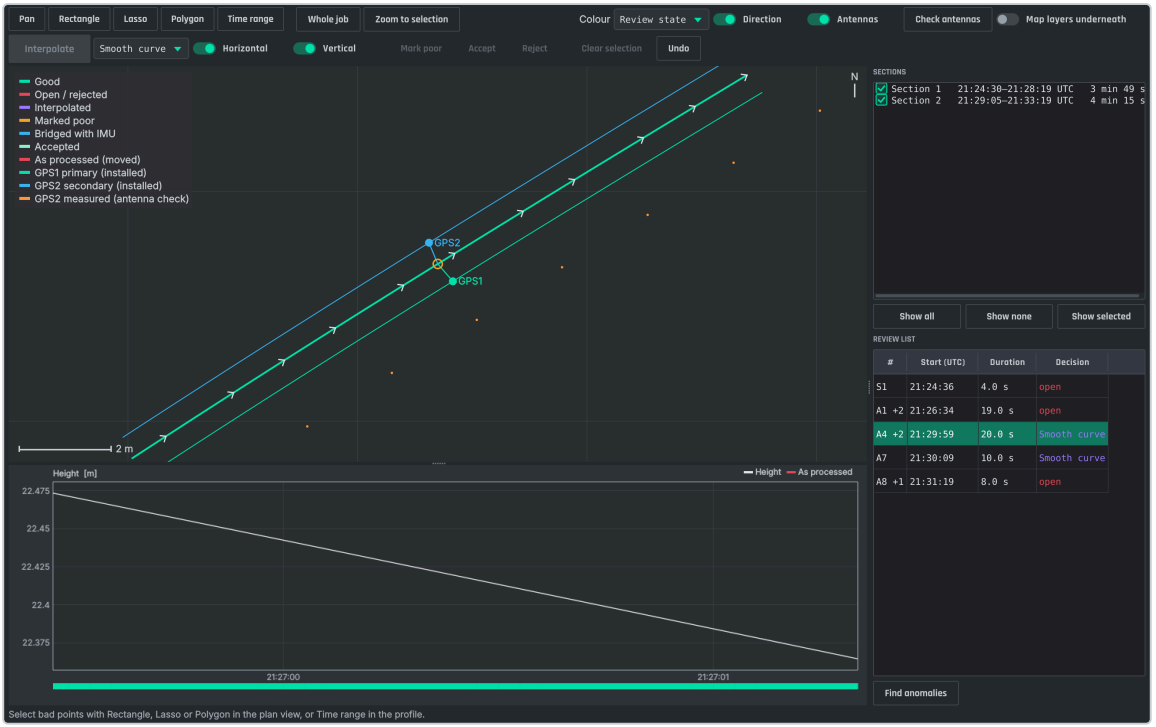


Figure 14.5 Direction chevrons, the antennas along the track and the vehicle at the clicked point, with the GPS2 positions measured by the antenna check (orange): they lie on the starboard side of GPS1, not on the blue GPS2 line, so the antennas were swapped.

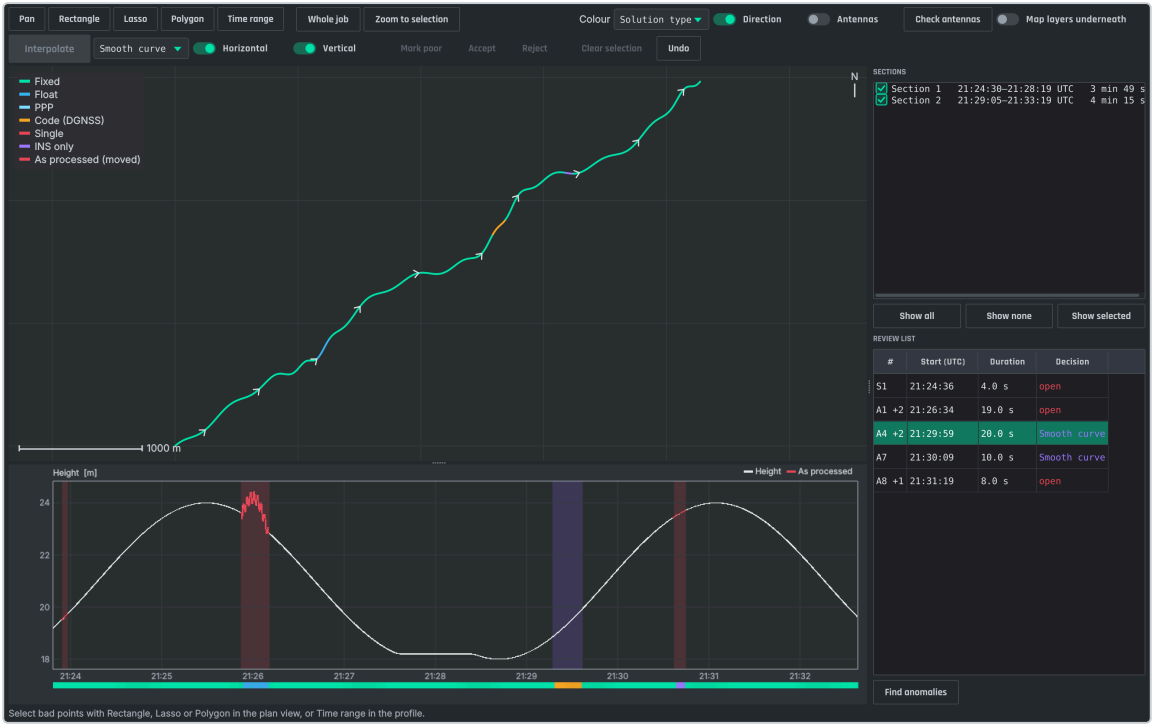


Figure 14.6 The whole job coloured by solution type: the height spike is interpolated (violet band), the second place is still open (red band).

Deciding

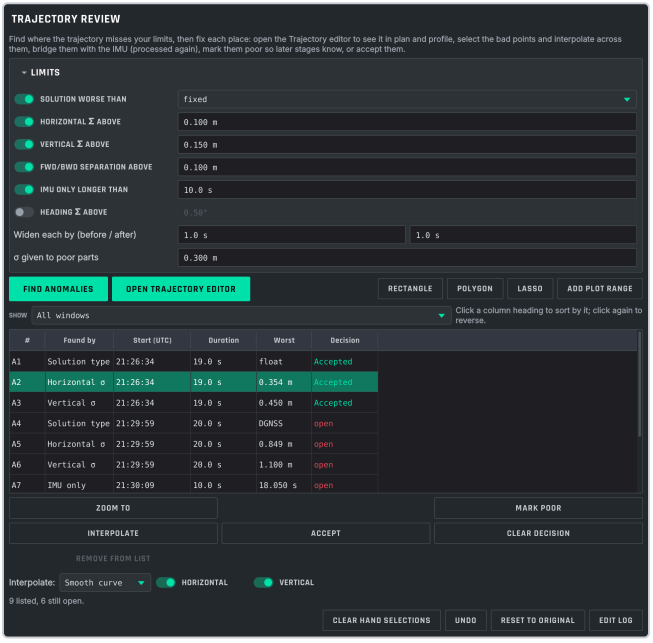


Figure 14.7 The same windows after the decisions: A1 marked poor, A2 accepted.

Select one row or several, then:

DECISION	EFFECT	APPLIES
BRIDGE WITH IMU	GNSS is not used in these windows; the IMU carries the trajectory, as through a tunnel. Use it where multipath or a wrong fix pulled the trajectory away. Tightly and loosely coupled only.	After **PROCESS AGAIN**; a note in amber says the exclusions are waiting, and the Review step reads *process again*
MARK POOR	The trajectory is kept, but its position σ is raised to at least **Σ GIVEN TO POOR PARTS**, in the SMRMSG file, on the Quality page and in the finalized trajectory, so later stages know not to trust it.	At once
INTERPOLATE	Replaces the positions between the good samples on both sides, as a *Smooth curve* or a *Straight line*, **HORIZONTAL** and/or **VERTICAL**, in any processing mode (see *The Trajectory editor*). For longer stretches of a coupled trajectory, **BRIDGE WITH IMU** usually follows the road better.	At once
ACCEPT	Looked at and fine as it is.	At once
CLEAR DECISION	The window is open again.	At once

Decisions never change the processed result itself: PosPro keeps an untouched copy and applies your decisions to it, so **UNDO** (the last change), **RESET TO ORIGINAL** (every decision removed, after a question) and processing again lose nothing. After a decision that changes the trajectory, the Quality page, the map and the plots are updated and the trajectory is finalized again. **EDIT LOG** lists every detection and decision with its time.

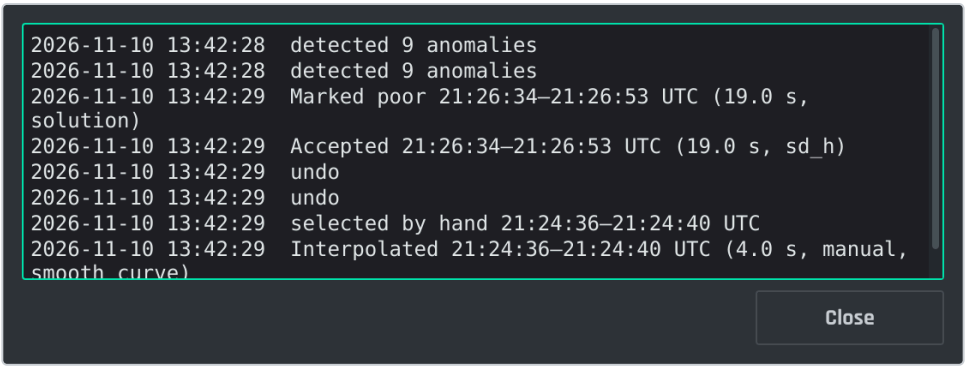


Figure 14.8 The Trajectory Review log.

> WINDOWS INSIDE WINDOWS Deciding on a window that lies inside a larger window that already has a decision adds a decision for the smaller window; the larger window keeps its own. To change a decision, select that window itself and decide again, or use `CLEAR DECISION`.

The Review step shows *find anomalies* before the first search, *N open* in amber while windows are undecided, *process again* while GNSS exclusions wait, and *N reviewed* in green when all are decided.

> NO BASE STATION? Choose `REAL-TIME SOLUTION (NO BASE STATION)` on the Processing page. PosPro then uses the trajectory the Certus computed in the field (RTK when it had corrections), and Trajectory Review cleans it the same way, with the Trajectory editor for the parts it got wrong.

15 Step 8 - Export

> WHY HERE Only for other software. AlignPro and CloudPro already read the trajectory PosPro finalizes every time it processes or applies a review edit.

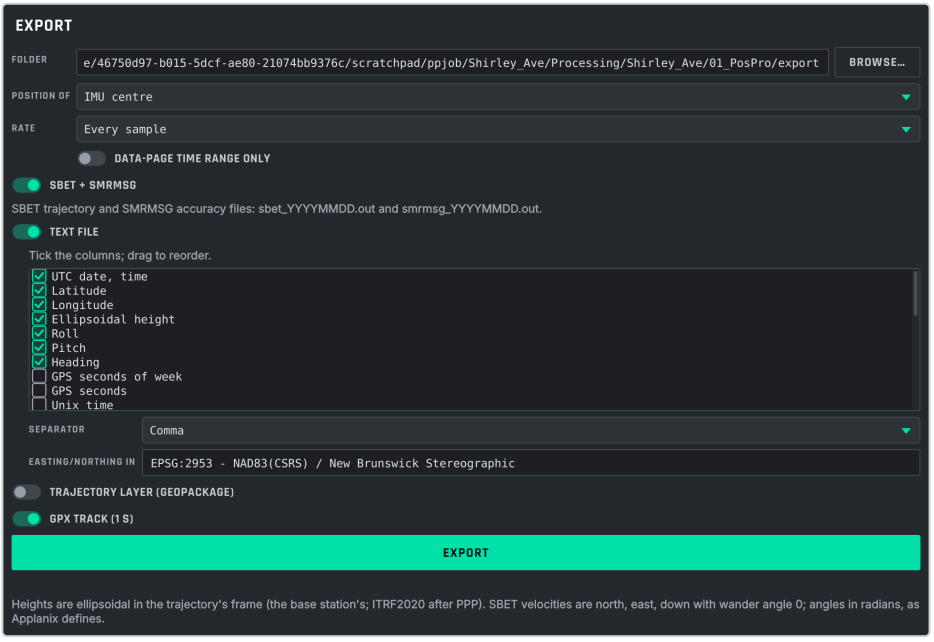


Figure 15.1 The Export page with SBET + SMRMSG, a text file with seven columns and a GPX track chosen.

SETTING	WHAT IT DOES
FOLDER	Where the files go; by default `01_PosPro/export` in the project. **BROWSE…** opens a folder picker.
POSITION OF	*IMU centre* (the Certus reference point), the output point from the Installation page (by its name, when its offset is not zero), or *GNSS antenna* (GPS1).
RATE	Every sample, 200, 100, 50, 10 or 1 Hz.
DATA-PAGE TIME RANGE ONLY	Export only the time range set on the Data page.
SBET + SMRMSG	The trajectory and its accuracy: `sbet_YYYYMMDD.out` and `smrmsg_YYYYMMDD.out`.
TEXT FILE	`<project>_YYYYMMDD.txt` with the columns you tick, in the order you drag them to: UTC date, time; GPS seconds of week; GPS seconds; Unix time; Latitude; Longitude; Ellipsoidal height; Easting; Northing; Roll; Pitch; Heading; Velocity N, E, D; Speed; σ North, East, Down; σ Roll, Pitch, Heading; Solution type; Satellites; Separation H, V. **SEPARATOR**: Comma, Tab, Space or Semicolon. **EASTING/NORTHING IN**: a transverse Mercator, UTM or stereographic system (EPSG:2953, NAD83(CSRS) / New Brunswick Stereographic, by default).
TRAJECTORY LAYER (GEOPACKAGE)	`<project>_trajectory.gpkg`: the Trajectory map layer (one point per second with its quality attributes).
GPX TRACK (1 S)	`<project>_trajectory.gpx`: the trajectory at the Certus reference point, one point per second, with ellipsoidal heights.

Press **EXPORT**; the files written are listed under the button, and the Export step turns green (*exported* and the time) until the trajectory is processed again. Every export also writes `<project>_export_note.txt`, which records the project, when and how it was processed, the frame and epoch, the base and its position source, the heights, the time system, the exported point and its offset, and the attitude frame. Keep it with the files.

File formats

FILE	CONTENT
`sbet_YYYYMMDD.out`	17 little-endian doubles per record: GPS seconds of week, latitude, longitude (rad), ellipsoidal height (m), velocity north, east, down (m/s), roll, pitch, heading (rad), wander angle (0), body accelerations, body angular rates.
`smrmsg_YYYYMMDD.out`	10 doubles per record: GPS seconds of week, σ north, east, down (m), σ velocity north, east, down (m/s), σ roll, pitch, heading (arc-minutes).
`<project>_YYYYMMDD.txt`	One header line with the column names, then one line per sample.

Heights are ellipsoidal, in the frame of the base station's position (for PPP: ITRF2020 at the observation date; for the real-time mode: the frame of the Certus real-time solution, that is of the RTK corrections used in the field). Apply the geoid in the point-cloud processing. SBET velocities are north, east, down with wander angle 0; angles are in radians. Use the same reference point in the point-cloud processing as in the export.

What PosPro hands to AlignPro and CloudPro

At the end of every processing run, and after every review decision that changes the trajectory, PosPro finalizes the trajectory in `01_PosPro/out/`. You do not need the Export page for AlignPro and CloudPro: they read these files.

FILE	CONTENT
`trajectory_sbet.out`	SBET at the Certus reference point (IMU centre), every sample
`trajectory_smrmsg.out`	SMRMSG for the same samples (σ raised where you marked parts poor)
`trajectory_qc.csv`	The 1 Hz quality table (position, attitude, velocity, σ, separation, solution type, satellites, gap, IMU biases)
`trajectory.json`	The description of the trajectory, written last (schema `dcs.trajectory/1`)

TRAJECTORY.JSON

FIELD	HOLDS
`schema`, `id`, `producer`, `created`, `project`	`dcs.trajectory/1`; a content id of the data files (the later stages use it to see whether they are out of date); PosPro and engine versions; time; project name
`built_from`	The captures (by fingerprint) and the shared settings the trajectory was made from
`files`	`sbet`, `smrmsg`, `qc`
`time`	System GPS, GPS week, leap seconds, start and end (GPS seconds), SBET time as GPS seconds of week
`frame`	Frame name and epoch (that of the base position); for a real-time trajectory, the frame of the Certus's RTK corrections; heights ellipsoidal
`reference_point`, `body_frame`	Certus reference point (IMU centre), offset 0, 0, 0; Certus aligned vehicle frame: X forward, Y starboard, Z down; roll + starboard down, pitch + bow up, heading + clockwise

`levers_m`	GNSS 1 and GNSS 2 offsets (with their source, and whether GNSS 2 was used), the Lidar offset from the capture's `hardware.json`
`rate_hz`	The rate of the finalized SBET, measured from its time stamps
`quality`	Method, positioning, % fixed, % IMU only, separation (95 %), largest horizontal and vertical σ
`base_station`	The base used
`review`	The Trajectory Review decisions (bridged with the IMU, reduced quality with its σ, interpolated, accepted windows) and the limits used
`attitude`	For GNSS-only trajectories only: attitude not observed (roll and pitch 0, heading from the direction of travel)

When you change something the trajectory depends on (the coordinate frame, the base station, the antenna offsets) or a capture changes, the PosPro box turns ⚠ **stale** in every plugin, and AlignPro and CloudPro show that their results are older than the trajectory. Process again to bring the chain up to date.

16 Marine jobs

A job whose captures are sonar recordings from the survey boat (`.clraw` of kind *sonar*, made by RemixPro from SonarView and the Reach files) opens on one page, **Marine trajectory**, instead of the eight Lidar steps.

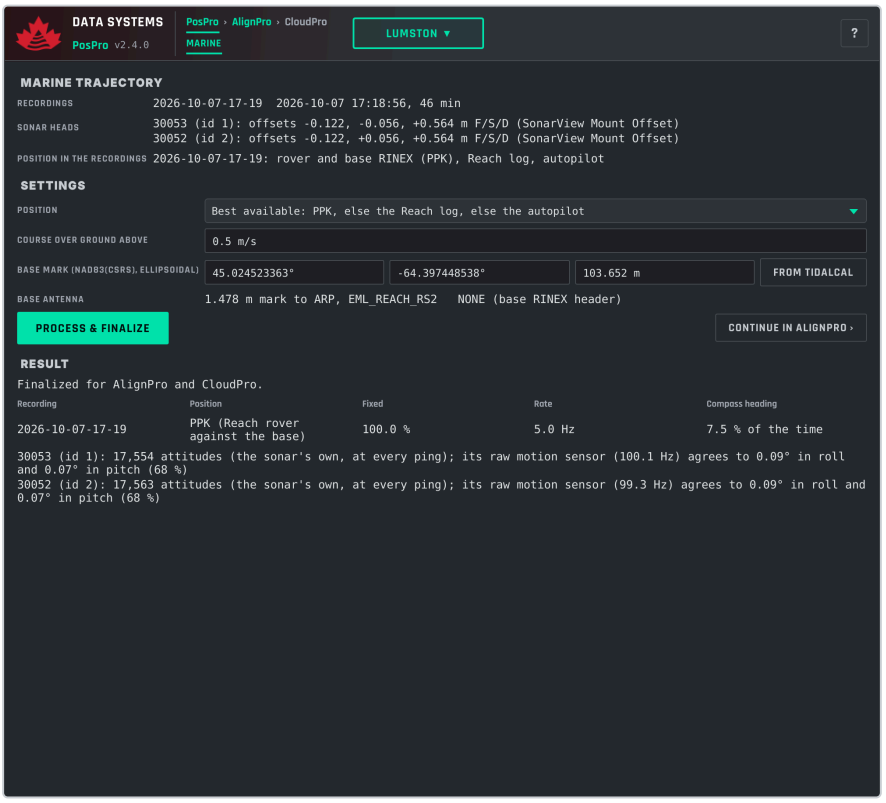


Figure 16.1 The Marine page after Process & finalize on the Lumston survey: PPK against the base in the recording, 100 % fixed; each head's own attitude at every ping.

What the page shows

- **Recordings**, their start and length, and the **sonar heads** with their offsets from the GNSS antenna (forward, starboard, down) and where those came from (SonarView's Mount Offset, a `sonar_offsets.json`, the TidalCal project).
- **Position in the recordings**: rover and base RINEX (PPK possible), the Reach real-time log (.LLH), the autopilot.

Settings

SETTING	MEANING
Position	*Best available* (default): PPK when the rover and base RINEX are in the recording, else the Reach log, else the autopilot. Or one of them.
Course over ground above	The boat's heading is its course over ground when it moves faster than this (default 0.5 m/s), and the autopilot's compass below. The page says how much of the time the compass was used.
Base mark	The surveyed base point in NAD83(CSRS), ellipsoidal height. `FROM TIDALCAL` takes the base from the TidalCal project. The antenna height is read from the base RINEX header (mark to the antenna reference point, as entered in Emlid Flow); PosPro adds the antenna's phase-centre offset.

Process & finalize

PosPro computes the boat's trajectory at the GNSS antenna (PPK in both directions, combined), its heading, and each head's attitude: the sonar's own fused attitude at every ping (packet 504), checked against its raw motion sensor (packet 505); the page shows how closely they agree. The trajectory and the attitudes are finalized for AlignPro and CloudPro. Then `CONTINUE IN ALIGNPRO ›` for the sonar patch test.

> **.CLRAW FROM REMIXPRO 1.2.0** RemixPro 1.2.0 read SonarView's Mount Offset Y as starboard; it is port. DCS Core 1.14.0 corrects those files when it reads them, so they need not be made again.

17 Troubleshooting

Messages

MESSAGE	WHAT TO DO
add the rover files (.anpp or .clraw) first / rover file not found: …	Add the files on the Data page; if the job folder moved, use **RAW DATA MOVED? RELOCATE THE JOB FOLDER…** .
no raw IMU (packet 28) with valid time was found / no raw satellite observations (packet 60) were found	The Certus did not record these packets. Only the real-time mode can run on these captures. Set the Certus outputs on the mapper for the next capture.
Read the rover files (Data). / Add a base station (Base stations).	Shown by **PROCESS** when a step before it is missing.
the base station … has no position yet	Enter or compute the base position on the Base page.
tight coupling needs a base station; for PPP choose "Loosely coupled"	PPP works with loose coupling or GNSS only.
no navigation (ephemeris) data: connect to the internet or add a RINEX navigation file on the Data page	Turn on **DOWNLOAD BROADCAST EPHEMERIS** with internet, or add a navigation file.
navigation file …: N orbits disagree with the INS's own and the IGS broadcast orbits … and are left out (…)	That file holds wrong orbits for those satellites. PosPro already left them out; take the file out of the base folder so it is not used again.
the solution ran away: its σ reached … km	Processing stopped: the trajectory cannot be used, and Trajectory Review cannot repair it. Check the warnings above it: a bad navigation file, a wrong base position (compare the PPP result with the RINEX header position or CSRS-PPP), or IMU data that did not arrive in full. Fix the cause and process again.
the largest σ is … m	A warning: look at the quality plots before using the trajectory.
broadcast ephemeris for … not downloaded (…)	A warning: PosPro goes on with the files you supplied and the rover's GPS ephemeris.
no precise orbit/clock found for …	Final products appear about two weeks after the day, rapid ones after about a day. Try again later, or use CSRS-PPP.
antenna calibrations (igs20.atx) not available	Base antenna phase-centre variations are not applied; connect to the internet once so the file is cached.
the base files … list different observation types; add them as separate base stations	Add each file as its own base.
the base file is only … h long	PPP is good to a few cm only after 2 h; use a longer file or CSRS-PPP.

CSRS-PPP needs the e-mail address you use as your NRCan user name	Enter it under **E-MAIL**.
the rover files hold no real-time solution (packet 20)	The real-time mode cannot run on these captures.
GPS2 could not be measured (no GPS2 observations, baseline did not fix)	Record a few minutes with both antennas in the open, or enter GPS2 by hand.
the lever-arm solver needs tight coupling / the filter did not report lever arms	Process tightly coupled with a base, with most of the drive fixed.
Easting/northing: choose a transverse Mercator, UTM or stereographic system	Choose such a system on the Export page.
process the project first	Export needs a processed trajectory.
… is being processed by … on … since …	Another QGIS is processing this stage of the project. Wait for it to finish.
…% of the IMU samples are missing (… gaps, the longest … s): the INS stream did not arrive in full	The recording has holes. PosPro bridges short gaps; for the next capture record with mapper 2.10.1 or later and lower the output rates (Recording settings).
the IMU's aligned frame is … deg (pitch) / … deg (yaw) off the vehicle	The INS alignment set in the Certus does not match how it is mounted. PosPro allows for it (the offset is estimated while driving); correct the alignment in the Certus so its real-time solution benefits too.
The processing engine is missing	Reinstall PosPro. On Windows, allow `pospro-engine.exe` if your security software asks.
The user guide is missing from this installation; reinstall PosPro.	Reinstall PosPro.

Results

WHAT YOU SEE	LIKELY CAUSE	WHAT TO TRY
Fixed below 75 %	Base far away (over 40 km), poor sky	A nearer base; **ATMOSPHERE** on Automatic or Always; a higher elevation mask
Large separation at the start or end only	Vehicle parked: heading cannot be resolved without motion	Drive a few turns before and after the survey; use both antennas
Large separation in a long gap	IMU drift both ways	Expected; check the σ plot over the gap; the wheeled-vehicle constraint and the rear axle offset help
Everything offset by the same amount	Base position from the RINEX header, or the wrong frame or epoch	Set the base position (CSRS-PPP)
Trajectory tilts or swings in turns	Wrong antenna offset or rear axle offset	Check the Installation drawing; solve the lever arms

18 Reference

Defaults of a new project

SETTING	DEFAULT
Frame	NAD83(CSRS), epoch 1997.0 (until a base position sets it)
Method / positioning / passes	Tightly coupled / PPK / forward and backward
Constellations, mask, ratio	GPS, GLONASS, Galileo, QZSS; 12°; 3.0
Ambiguities, frequencies, atmosphere, SNR weighting	Fixed; dual; automatic; on
Wheeled vehicle constraint, zero velocity, rate	On; on; IMU rate
Alignment, speed, gate, max base gap	Automatic; 4 m/s; χ^2 9; 60 s
IMU, vehicle	Certus Evo; Van (or the vehicle recorded in the capture)
Export	IMU centre, every sample, SBET + SMRMSG
Review limits	See Trajectory Review

Folders

PATH (IN PROCESSING/<NAME>/01_POSPRO/)	HOLDS
`pospro.json`	PosPro's settings, bases, results summary and review decisions
`work/extract/`	The data read from the captures (IMU, observations, real-time track, coverage, hardware record)
`work/base/`	Downloaded and prepared base files, base PPP
`work/gnss/`, `work/ins/`	GNSS solutions, inertial passes and their settings
`work/results/`	The combined trajectory, its quality table and their untouched copies for Trajectory Review
`out/`	The finalized trajectory for AlignPro and CloudPro
`export/`	Exports (default folder)

Recording settings

The first field test (3 Nov 2026, mapper 2.9.0) lost 31 % of the Certus stream in 357 gaps of up to 1.4 s. PosPro bridges short gaps, and the Data step reports how much IMU data is missing, but a clean recording is always better. Two causes add up: the mapper's computer, busy recording the Lidar, could drop network packets, and the Certus stream then stalls and the Certus throws data away while it waits (mapper 2.10.1 reduces that load); and the outputs asked for about 69 KB/s, close to what came through. When the Data step reports missing IMU samples, record with mapper 2.10.1 or later and lower the output rates:

PACKET	WHAT POSPRO USES IT FOR	SETTING
20 System state	time for every IMU sample, real-time solution	5 ms (200 Hz), keep

28 Raw sensors	the IMU data	5 ms (200 Hz), keep
60 Raw satellite data	GNSS observations of both antennas	1000 ms (matches a 1 s base); 500 ms if the check below stays clean
61 Ephemeris; 185, 196 offsets	orbits; installation and dual-antenna offsets	as the Certus sends them
29 Raw GNSS fix	not used for post-processing	100 ms or off
26 Euler standard deviation	not used	off

With these settings the stream is about 37 KB/s. Check after the next drive: the Data step should report under 1 % missing IMU samples.

Field practice that improves the trajectory

- Record raw IMU and raw GNSS of both antennas on the Certus with GLONASS on (it helps the fix rate), within what the port can carry (Recording settings).
- Log a base at 1 s, within 20 km, for the whole drive plus 10 minutes either side. Park 1–2 minutes, then drive a few turns at the start and the end; this resolves heading and IMU biases for both passes.
- Measure the antenna offsets to the millimetre in the Certus aligned frame, and the rear axle to 5 cm; then check them with the lever-arm solver.

19 Version history

VERSION	CHANGES
2.4.0	**Marine jobs:** a job of sonar recordings (made by RemixPro) opens on a single Marine page: the boat's position by PPK, the Reach log or the autopilot, the heading from the course over ground, each sonar head's own attitude, and Process & finalize for AlignPro and CloudPro (Marine jobs). Needs DCS Core 1.14.0.
2.3.0	**Recording holes:** the finalized trajectory records where it has no data, and Trajectory Review says so at the top. Needs DCS Core 1.13.0.
2.2.1	**Opened from CloudPro:** `OPEN IN POSPRO` in CloudPro's LINE SEPARATION tab brings up Trajectory Review with the point cloud rows shown and the chosen problem area selected. Needs DCS Core 1.12.0.
2.2.0	**Problem areas from the point cloud:** the places where CloudPro's line separation check found one line's points off the others are listed in Trajectory Review (C rows) and drawn in the Trajectory editor, with the line and the time, to decide on like any other window (Problem areas from the point cloud).
2.1.0	The new DCS header: the chain and PosPro's phases (Prepare › Process › Finalize, formerly Finish) as breadcrumbs, with the guidance in their tooltips instead of a sentence between Back and Next. The window is always a window of its own (no docking); `RESET WINDOW` is in the project menu. Messages say *finalize* instead of *publish*. Needs DCS Core 1.11.0.
2.0.5	The GNSS processing starts afresh after a break of more than 30 s in the rover data, for example between two survey lines. Before, the forward pass could stay float for the whole of the next line. Quality adds a finding with the forward/backward agreement where both passes are fixed, and the share of the drive where only one pass is fixed (the combined result follows the fixed pass there, so the overall separation overstates the error).
2.0.4	Inertial processing starts again from the Certus attitude and position after an IMU gap longer than 5 s, for example between two survey lines where the van turns round. Before, the heading could come out about 180° wrong on one line, so GPS1 and GPS2 appeared to cross and the AlignPro patch test could not be solved: process jobs with more than one line again. The Data step warns when the recording lost IMU data.
2.0.3	A job processed before 2.0.1 with an output point set is flagged when you open it: the Processing step reads *made before 2.0.1 with an output point: process again* and a message explains why (its published trajectory makes AlignPro and CloudPro apply the Lidar offset twice). Press `PROCESS` to publish it again. New projects are named after the date of the first capture, for example *2026-11-03_1* (the next number when that name is taken).
2.0.2	**System profiles** (section System profiles): `SYSTEM PROFILE…` on the Installation page saves the checked antenna offsets, rear axle, IMU and vehicle as the profile of the system that recorded the job, and PosPro offers a matching profile when you read the rover files of a later job. AlignPro keeps the Lidar mounting in the same profile. Needs DCS Core 1.7.0.
2.0.1	Fixes from a full audit. **The published trajectory is always at the IMU centre**: with an output point set (for example by `USE OFFSETS FROM THE CAPTURE`) it was at that point but labelled as the IMU centre, so AlignPro and CloudPro applied the lidar offset a second time – process again (or press `PROCESS`) to republish. The frame is worked out for each run (PPP: ITRF2020 at the observation date; PPK: the base position's frame). Tight coupling starts only from GNSS-only solutions made with the same base, position and settings. Review edits made while another job runs are applied when it ends, and quick decisions in a row are applied together. The real-time mode honours the time range. Buttons stay off until they can work and while a job runs; the Export page keeps your choices; Processing reads *settings changed* when its inputs change; Review reads *done* when nothing is over the limits. Downloads are streamed to disk, retried once and refreshed when fetched on the same day. Engine 2.0.1: a damaged record in a .clraw file is reported and skipped instead of ending the read; dual-antenna jobs prepare much faster. On the Review page `EXCLUDE GNSS` is now `BRIDGE WITH IMU` and `PLOT VIEW` is `ADD PLOT RANGE`, as in the editor.

2.0.0	A guided workflow in three phases: **Prepare** (Data, Installation, Base), **Process** (Processing, Quality, Calibration), **Finish** (Review, Export). Installation now comes before Base, so the antennas are checked before anything is processed, and Calibration before Review, because solving the lever arms means processing again (see *The workflow and why it is in this order*). The Base step reads *not needed* in real-time mode. This guide rebuilt: how a trajectory is made, the workflow, best practices, a *Why here* note for every step, workflow diagrams, contents and an index, and screenshots of this version.
1.8.0	`CHECK ANTENNAS` (Installation page and Trajectory editor): measures where GPS2 really was relative to GPS1 while driving straight, using the direction of travel rather than the Certus heading, and says whether the antennas match the installation or look swapped; one click then swaps the offsets and aligns from motion. Trajectory editor: `DIRECTION` chevrons along the track, `ANTENNAS` (GPS1 and GPS2 drawn along the track, and the vehicle at the point under the mouse) and the measured GPS2 positions. The QGIS map's trajectory and real-time lines show arrowheads in the direction of travel.
1.7.0	**Trajectory editor**: the trajectory in plan and in a height profile side by side, linked both ways – hover or click in one and the other shows the same moment, select bad points in either (rectangle, lasso, polygon, time range) and fix them there. `INTERPOLATE` now works in every processing mode, as a *smooth curve* (follows bends) or a *straight line*, horizontally and/or vertically, also at the start or end of the trajectory. New decision `REJECT` (bad, to fix later). See *Trajectory Review*.
1.6.3	When DCS Core is updated while QGIS is running, PosPro says to restart QGIS instead of asking for an update.
1.6.2	Installation: `IMU MOUNTING CORRECTION` – the pitch and yaw of the vehicle against the IMU, as a start for the estimate or held fixed, and `USE THE VALUES FROM THE LAST PROCESSING`. Trajectory Review: a `SHOW` filter and sorting by any column.
1.6.1	Navigation files next to the base observations, and files you add, are checked orbit by orbit against the Certus's own orbits and the IGS broadcast orbits before processing; orbits that disagree are left out and a warning names the file and the satellites (see *Data*). Processing stops with a clear error when the solution runs away (σ above 1 km) instead of handing over an unusable trajectory. Trajectory Review: `REMOVE FROM LIST` and `CLEAR HAND SELECTIONS`; one drawing on the map adds one window (or a few), not one per piece; hand selections from before are taken off the list when you process again, and the anomalies are found again on the new trajectory with your limits.
1.6.0	Guided steps: a `‹ BACK` / `NEXT ›` bar under the pages says whether each step is done and what to press, and `CONTINUE IN ALIGNPRO ›` on the last step once the trajectory is published (see *The PosPro window*). The version number is shown after the plugin name in the header. Fix (DCS Core 1.5.1): QGIS 4 on Windows no longer closes when you click in a floating window. Needs DCS Core 1.6.0.
1.5.3	Fixes from the first real field test: a cycle slip on a signal that returns after a short dropout is caught (it could pull a fixed solution metres off); short losses of IMU data are bridged and the Data step reports how much is missing; the offset between the vehicle and the IMU's aligned frame is estimated while driving and reported; tightly coupled passes start from the fixed GNSS-only position. On the test drive: under 35 % fixed before, 99.8 % now. New: Recording settings.
1.5.2	Fix: no more "unclosed file" warnings in the QGIS log after processing.
1.5.1	Mapper 2.10 projects: a new project offers the navigation recordings (recommended) or the survey lines; rover files are read in recording order; a navigation recording's hardware record is completed from a survey line. Needs DCS Core 1.4.0.
1.5.0	The same header as AlignPro and CloudPro: logo, the plugin name, the workflow boxes beside it, the project box on the right, then the Window button and the user guide button. The window title is just *PosPro*. The vehicles are the mapper's (Van, Truck, Car) and set the wheel constraints; the separate motion-constraints list is gone (older projects keep what they processed with). Captures without raw IMU take their time span from the real-time solution, and a real-time trajectory is published with the frame of the Certus RTK corrections and its own rate. This guide, rewritten with screenshots of this version. Needs DCS Core 1.3.0.
1.4.0	Named just PosPro. Opens in DCS Core's standard window: floating the first time, docked only when you choose it from the Window menu; dragging never docks it. It comes back as you left it, also after QGIS restarts; `RESET WINDOW` returns to the default. The quality plots open in the same kind of window. Neutral graphite look with lighter view backgrounds.

1.3.x	Trajectory Review (Review). The real-time mode (the Certus solution, no base station). GPX export. The user guide is installed with the plugin.
1.2.x	Part of the DCS chain (needs DCS Core): projects in the job folder, the trajectory published for AlignPro and CloudPro, workflow boxes with out-of-date results, relocating a moved job folder, saving the QGIS project in the project folder. Older `.pospro.json` projects still open.
1.1.x	Tight coupling with both antennas, atmosphere estimation, signal-strength weighting, lever-arm solver, base-station finder, Installation filled from the capture, data coverage bars, **MEASURE GPS2 FROM THE DATA**.

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